

TAXATION THROUGH NEGOTIATION

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Governments can extend their tax enforcement capacity by deputizing firms to remit tax, report information about taxable transactions, or both. While these actions are usually mandated, firms occasionally volunteer to assist the government in collecting taxes, typically to forestall unwelcome regulation. But do such agreements increase tax revenue? And under what circumstances should governments make the necessary compromises to implement them? I study both questions in the context of Airbnb's Voluntary Collection Agreements (VCAs): bilateral contracts between Airbnb and various U.S. cities which obligated the platform to remit hotel taxes on behalf of its proprietors. Using public records and Freedom of Information Act requests, I construct an original dataset of hotel tax revenue and use causal inference methods to estimate revenue increases of 12-15% following VCA adoption, corresponding to approximately 85% of projected tax liability.

Despite their efficacy, I also document many instances in which signatory cities struggle to enforce non-tax regulations without access to individual proprietor information—data they agreed not to pursue as a condition of the VCA. On balance, I argue that, while VCAs will likely remain the only realistic path to securing tax revenue in some contexts, they can hamper enforcement of non-tax regulations and may undermine revenue collection in the long run. I conclude by suggesting strategies for governments to leverage better terms when negotiating VCAs.

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INTRODUCTION

In 2014, Airbnb’s popularity suddenly exploded. By offering an alternative to traditional hotels, the platform went from a single listing to more than 30 million guests in just six years.¹ But Airbnb did more than disrupt the travel industry—it disrupted municipal tax collection. The emergence of short-term rental platforms multiplied the universe of potentially taxable entities from hundreds of hotels to thousands of anonymous private hosts, without any corresponding expansion of enforcement capacity.² This shift exposed a critical weakness in local governments’ ability to collect taxes from novel business models and led to the emergence of controversial Voluntary Collection Agreements (VCAs) between city governments and Airbnb.

VCAs are public contracts Airbnb voluntarily entered into, obligating the corporation to assess and remit hotel tax liability generated on its platform.³ In exchange, cities made a variety of concessions—agreeing not to pursue back taxes, for example, or liberalizing short-term rental regulation.⁴ I argue that the most important of these concessions was cities’ agreement to accept

¹ See Airbnb, Inc., Registration Statement (Form S-1) 1, 131 (Nov. 16, 2020) (documenting the company’s founding in 2007, when “two Hosts welcomed three guests to their San Francisco home,” and its growth to over 825 million cumulative guest arrivals by September 2020); Morgan Brown, *The Making of Airbnb*, BOS. HOSP. REV. (Jan. 8, 2016) (noting that the platform had ten million guests by spring 2014 and nearly doubled its bookings year-over-year).

² See Paris Martineau, *Inside Airbnb’s ‘Guerrilla War’ Against Local Governments*, WIRED (Mar. 20, 2019) (describing how Airbnb’s more than six million listings outpace the top six hotel chains combined, creating enforcement challenges for municipalities that lack the infrastructure to identify and tax thousands of anonymous hosts); Josh Bivens, *The Economic Costs and Benefits of Airbnb: No Reason for Local Policymakers to Let Airbnb Bypass Tax or Regulatory Obligations*, ECON. POL’Y INST. (Jan. 30, 2019).

³ See, e.g., Voluntary Collection Agreement for Collection of Transient Guest Tax, Kan. Dep’t of Revenue & Airbnb, Inc. (on file at <https://www.ksrevenue.gov/pdf/airbnbvca.pdf>) (requiring Airbnb to “collect and remit” applicable transient guest taxes on behalf of hosts).

⁴ See Martineau, *supra* note 2 (noting that VCAs “prohibit cities from attempting to collect back taxes” and that in Portland, “city officials legalized home-sharing and lowered the registration fee for short-term rentals around the same time Airbnb agreed to add a 11.5 percent occupancy tax on each booking”); *Airbnb Targeted by States Suspecting Tax-Collection Shortfalls*, Bloomberg Tax (Feb. 4, 2019) (describing VCA terms that prevent governments from learning host names or addresses).

lump-sum remittance from Airbnb without identifying information about the hosts who generated the liability. Were these trade-offs worth it? That question has two parts: how effective were VCAs at delivering tax revenue, and what did cities give up in exchange?

To understand the importance of VCAs, it is essential to place them in the larger dynamics of intermediary tax enforcement. Firms⁵ assist the government in collecting tax through two distinct actions: remitting tax and reporting information about taxable transactions. These actions are usually performed together—an employer both remits estimated income tax liability on behalf of an employee⁶ and files Form W-2, which reports total wages paid.⁷ But they can also be decoupled. A firm that hires an independent contractor, for example, must file Form 1099-MISC reporting how much the contractor was paid, but is not required to withhold and remit income tax on the contractor's behalf.⁸ Arrangements in which a firm reports but does not remit are well studied. Far less understood are arrangements in which the firm remits but does not report—such as Airbnb VCAs. This decoupling of remittance from information will become increasingly relevant as under-resourced state and local governments seek to collect taxes from disruptive platform firms.

⁵ I use the term “firms” in the economic sense, i.e. broadly referring to business organizations. Thus construed, partnerships and non-profits with reporting obligations would both be considered firms.

⁶ Note that, while the actions of withholding and remitting are distinct, in practice there is no withholding without remittance. Therefore, I use the term remittance to refer to them jointly.

⁷ I.R.C. § 3402(a)(1) (requiring every employer making a payment of wages to “deduct and withhold” a tax from such wages); I.R.C. § 6051(a) (requiring every employer subject to withholding under § 3402 to furnish each employee a written statement reporting total wages paid during the calendar year). *See generally* Joel Slemrod, *Tax Compliance and Enforcement*, 57 J. ECON. LITERATURE. 904, 921–25 (2019) (surveying the literature on withholding and third-party reporting as enforcement mechanisms).

⁸ I.R.C. § 6041(a) (requiring any person engaged in a trade or business who makes payments of \$600 or more to another person to report such payments to the Secretary); I.R.C. § 3401(a), (c) (defining “wages” subject to withholding as remuneration paid by an “employer” to an “employee,” thereby excluding payments to independent contractors from the withholding obligation). Nonemployee compensation was historically reported on Form 1099-MISC; beginning in tax year 2020, it is reported on Form 1099-NEC. *See* I.R.S., Instructions for Forms 1099-MISC and 1099-NEC (Rev. Dec. 2024).

This Article makes two contributions—one positive, one normative. First, I leverage a natural experiment in the staggered adoption of VCAs to estimate their effect on city hotel tax revenues. I construct a dataset of monthly municipal hotel tax revenues using public records and Freedom of Information Act (FOIA) requests. Using these data, I estimate a difference-in-differences model that exploits the staggered timing of VCA adoption across cities: because different cities signed agreements with Airbnb at different times, later adopters serve as controls for earlier ones. I find that VCAs increased hotel tax revenues by approximately twelve to fifteen percent—a statistically significant and economically meaningful effect that materialized immediately upon adoption. This is, to my knowledge, the first causal evidence on the revenue effects of voluntary collection agreements between a platform firm and municipal governments.

My second contribution is normative: I assess the conditions under which cities should sign VCAs. For this, I survey the experience of prominent cities that signed agreements. I find that the most consequential concession cities made was agreeing to accept aggregate tax remittance from Airbnb without identifying information about the hosts who generated the liability. This forfeiture of individual-level data left cities unable to audit Airbnb's remittances, enforce zoning and consumer safety regulations against non-compliant hosts, or track hosts who migrated to competing platforms. However, I argue that VCAs remain defensible under certain conditions—particularly for tourism-dependent cities with limited enforcement capacity facing hostile state preemption of local regulatory authority—and propose strategies, including collective bargaining among cities and standardized data-sharing requirements, to address the structural asymmetries that weakened cities' negotiating positions.

The Article proceeds as follows. Part I offers a simple framework for understanding VCAs in the wider context of intermediary tax collection. Part II presents the case study of Airbnb VCAs: I first discuss their structure and motivating logic, then explore the economic, legal, and political characteristics of Airbnb's business model that make it harder to regulate than traditional hotels. With this background, I turn to estimating VCAs' effects on tax revenue and examining indirect consequences, such as whether VCAs cause hosts to switch platforms to evade

taxation. After documenting the backlash to VCAs by some cities, I discuss the conditions under which signing one may still be the best course. Finally, Part III proposes concrete strategies for municipal regulators to better meet future platform entry into new industries.

This Article contributes to two broader themes in tax policy. The first is the relationship between state capacity and optimal tax design. A substantial literature recognizes that the “best” tax system is not one defined in the abstract, but one suited to the enforcement capabilities of the government implementing it. Governments that have invested in fiscal capacity—robust auditing infrastructure and comprehensive third-party reporting networks—can sustain high tax-to-GDP ratios with minimal evasion.⁹ But where such infrastructure is absent, governments rationally adopt second-best policies: taxing turnover rather than profits because it is harder to evade,¹⁰ relying on consumption taxes over income taxes because only formal-sector firms generate an enforceable paper trail,¹¹ or ceding enforcement authority to private intermediaries whose cooperation comes at a price. VCAs belong in this last category. Cities signed agreements with Airbnb not because forgoing individual taxpayer data was desirable, but

⁹ Timothy Besley & Torsten Persson, *PILLARS OF PROSPERITY: THE POLITICAL ECONOMICS OF DEVELOPMENT CLUSTERS* (2011) (modeling fiscal capacity as a state investment that develops in clusters alongside legal capacity, and arguing that countries with cohesive political institutions invest more in the infrastructure needed to sustain effective tax systems); Henrik Jacobsen Kleven, *How Can Scandinavians Tax So Much?*, 28 J. ECON. PERSP. 77 (2014) (demonstrating that Denmark’s near-zero evasion rate is attributable to comprehensive third-party information reporting, which covers approximately 95% of taxable income and renders enforcement largely self-sustaining).

¹⁰ Michael Carlos Best, Anne Brockmeyer, Henrik Jacobsen Kleven, Johannes Spinnewijn & Mazhar Waseem, *Production Versus Revenue Efficiency with Limited Tax Capacity: Theory and Evidence from Pakistan*, 123 J. POL. ECON. 1311 (2015) (finding that governments with weak enforcement capacity rationally adopt production-inefficient tax instruments, such as turnover taxes, that reduce evasion by 60–70% relative to profit taxes).

¹¹ Roger Gordon & Wei Li, *Tax Structures in Developing Countries: Many Puzzles and a Possible Explanation*, 93 J. PUB. ECON. 855 (2009) (arguing that developing countries’ reliance on consumption taxes and tolerance of informality reflect a rational response to the constraint that governments can effectively tax only firms that use the formal financial sector, which generates an enforceable paper trail).

because their limited capacity to identify hosts and enforce compliance against a global platform left few realistic alternatives. In this sense, VCAs are a domestic analogue to the capacity-constrained tax bargains documented in the developing-country context.¹²

The second theme is the tension between tax and non-tax regulatory objectives. Governments do not merely collect revenue; they simultaneously enforce zoning, consumer safety, and housing regulations—and instruments chosen to advance one objective can undermine another. At the federal level, states that legalized cannabis generate billions in tax revenue from a substance that remains Schedule I, yet federal prohibition prevents cannabis businesses from deducting ordinary expenses and limits their banking access, pushing operators toward cash transactions that erode the paper trail on which tax enforcement depends.¹³ At the local level, cities offer property tax abatements to attract development, only to find these instruments redirect revenue away from public services while working at cross-purposes with affordable housing goals.¹⁴ VCAs fit squarely within this pattern. By accepting anonymized tax remittance from Airbnb in lieu of

¹² Deborah Bräutigam, Odd-Helge Fjeldstad & Mick Moore, *TAXATION AND STATE-BUILDING IN DEVELOPING COUNTRIES: CAPACITY AND CONSENT* (2008) (examining how the process of building tax systems in developing countries simultaneously shapes governance patterns and the fiscal social contract between citizens and the state).

¹³ In 2024 alone, states with legal adult-use cannabis markets collectively generated over \$4.4 billion in tax revenue from cannabis sales. *See Cannabis Tax Revenue & Nationwide Cannabis Tax Policy Blueprint*, TAX FOUND. (2025), <https://taxfoundation.org/research/all/state/cannabis-tax-revenue-reform/>. Yet Section 280E of the Internal Revenue Code prohibits businesses trafficking in Schedule I or II controlled substances from deducting ordinary business expenses, effectively imposing punitive tax rates on legal cannabis operators. *See* 26 U.S.C. § 280E (2018).

¹⁴ *See* Jenny Schuetz, *To Improve Housing Affordability, We Need Better Alignment of Zoning, Taxes, and Subsidies*, BROOKINGS INST. (Jan. 17, 2020), <https://www.brookings.edu/articles/to-improve-housing-affordability-we-need-better-alignment-of-zoning-taxes-and-subsidies/> (documenting how misalignment between local tax incentives, zoning, and housing subsidies exacerbates affordability problems); *see also* Edward L. Glaeser & Joseph Gyourko, *The Impact of Zoning on Housing Affordability* (Nat'l Bureau of Econ. Rsch., Working Paper No. 8835, 2002) (finding that restrictive zoning is closely linked to elevated property prices, reducing housing supply while preserving property tax bases for existing homeowners).

individual host data, cities secured revenue but surrendered the information they needed to enforce zoning, monitor safety compliance, and track hosts across competing platforms—trading regulatory capacity for revenue in ways that proved difficult to reverse.

I. DEPUTIZING FIRMS IN TAX ENFORCEMENT

This section provides an overview of the actions firms take when assisting the government in tax enforcement. This framework provides a conceptual basis for understanding the defining feature of a VCA: **aggregate remittance** of tax funds by firms. First, I introduce key terms—report and remit—and juxtapose them in a two by two matrix. Next, I provide examples of familiar tax systems characterized by each quadrant of the matrix and discuss the implications for compliance.

A. *Dual Functions: Remittance and Reporting*

Consider the actions that must be taken for an incurred tax liability to become government revenue. When a transaction occurs between two parties, it is usually the statutory assignee, or the party designated by the government in law as the one on whom the tax is levied¹⁵ who must facilitate tax payment by performing three

¹⁵ Tax statutes typically levy a tax on either the "demand side" or the "supply side" of the market. The distinction refers to the base to which the rate is applied. Generally, if the tax is levied on the supplier, they cannot include the tax as a line item on their invoice to the consumer or insinuate that they are authorized to collect money on behalf of the government. Put another way, the supplier must remit the tax out of the final price paid by the consumer. See Joel Slemrod, *Does It Matter Who Writes the Check to the Government? The Economics of Tax Remittance*, 61 NAT'L TAX J. 251, 255–60 (2008). In contrast, when the statutory assignee is on the demand side, the tax is added on top of the listed price. The statutory assignee is generally legally liable for the tax—though, particularly when the tax is levied on the demand side, joint liability is frequently imposed. While public economics has long theorized that key economic outcomes will be the same regardless of which side is subject to the levy, recent empirical evidence suggests circumstances where the statutory

distinct actions. First, they must assess the tax at the time of the transaction by identifying the transaction as taxable and numerically calculating the liability amount.¹⁶ Second, they must file or *report* the tax liability by providing the government with basic information about how the tax liability was generated.¹⁷ Finally, they must then transfer the assessed liability to the government, or *remit*, at a regular time interval—monthly, quarterly, annually—by physically dispatching money to the government, such as by writing and sending the tax authority a check or depositing money electronically into a government account.¹⁸ The last two actions of reporting and remitting usually occur at the same time, such as when an individual files their income tax return (reports) and pays any outstanding income tax liability (remits) but, as we will see, this is not necessary.¹⁹

This sequence becomes more complicated when the government requires that a party *other* than the statutory assignee facilitate tax compliance by assuming responsibility for one or more of these actions.²⁰ Other parties may be tasked with assessment, withholding of tax liability from the statutory assignee at the point of transaction²¹ and remittance of this

assignment does influence these outcomes. *See, e.g.,* Wojciech Kopczuk, Justin Marion, Erich Muehlegger & Joel Slemrod, *Does Tax-Collection Invariance Hold? Evasion and the Pass-Through of State Diesel Taxes*, 8 AM. ECON. J.: ECON. POL'Y 251 (2016).

¹⁶ *See* N.Y. State Dep't. of Tax'n. & Fin., *Publication 750: A Guide to Sales Tax in New York State* 22, 24, 31 (2015) (explaining that vendors must determine whether sales are taxable and compute the tax due by applying the applicable state and local rate to the amount of the sale).

¹⁷ *See* N.Y. State Dep't. of Tax'n. & Fin., *Filing Requirements for Sales and Use Tax Returns*, Tax Bulletin ST-275 (updated Nov. 20, 2015) (explaining that a sales-and-use-tax return reports gross sales, exempt and nontaxable sales, taxable sales, taxable purchases, credits, and taxes collected or required to be collected).

¹⁸ *See* N.Y. State Dep't. of Tax'n. & Fin., *supra* note 16 at 4, 33–35.

¹⁹ *See* I.R.C. §§ 6072(a), 6151(a) (requiring calendar-year individual income-tax returns to be filed by the statutory deadline and generally requiring the tax shown on a return to be paid at the time and place fixed for filing the return).

²⁰ Leandra Lederman, *Statutory Speed Bumps: The Roles Third Parties Play in Tax Compliance*, 60 STAN. L. REV. 695, 697–98 (2007).

²¹ This is sometimes referred to as “withholding at the source.”

amount to the government.²² Or they might be required to submit information relevant to individual tax liability which enables the government itself to enforce against the statutory assignee (reporting).²³ Not uncommonly, firms are required to do both.²⁴ As I will discuss in the remainder of this Part, the type of assistance rendered has a significant effect on compliance.

B. A Simple Tax Enforcement Framework

We can think of these two actions—remitting and reporting—as two axes of a simple two by two matrix that characterizes all tax regimes. I briefly discuss each quadrant, starting with a tax system in which there is both reporting and remittance and proceeding counter clockwise.

²² Lederman, *supra* note 20, at 697–98; Wei Cui, *Taxation Without Information: The Institutional Foundations of Modern Tax Collection*, 20 U. Pa. J. Bus. L. 93, 107–108 (2018).

²³ Lederman, *supra* note 20, at 697–98; IRS, *About Form W-2, Wage and Tax Statement* (Jan. 29, 2024), <https://www.irs.gov/forms-pubs/about-form-w-2>.

²⁴ Lederman, *supra* note 20, at 697.

Figure 1. The Compliance Matrix

	Remittance	No Remittance
Reporting	E.g. W-2 Wages 98%	E.g. 1099-MISC Income 60-70%
No Reporting	E.g. QI Program; VCAs ?	E.g. State Use Tax <15%

Notes. This figure consists of a two by two matrix, juxtaposing the two roles that firms perform when assisting the government in tax enforcement. Within each cell there is an example of a corresponding tax regime, and, printed in blue beneath the example, the estimated voluntary compliance rate. Each of this examples is treated in more detail, *supra* in Section I.B.

Information Reporting and Tax Remittance. When the tax authority requires a party other than the statutory assignee to remit and report, it is frequently the party opposite the statutory assignee in a taxable transaction.²⁵ For example, in a typical retail transaction subject to sales tax, the statutory assignee is the consumer, but almost all state governments require any retailer with nexus to assess, report, and remit the tax on the consumer's behalf.²⁶ Importantly, joint tax liability is frequently extended to those that remit and report—meaning that if they do not remit the tax liability from the statutory assignee, and the statutory assignee does not independently self-assess and remit it, they may be required to pay the government some or all of the liability in the assignee's stead.²⁷

Why does the government impose these burdens on the retailer? One reason might be that retailers incur fewer costs or frictions than consumers in facilitating tax payment. The retailer is already tallying the prices of all items at the register, knows

²⁵ *Id.*, at 697, 699–700.

²⁶ New York State Dep't. of Tax'n. & Fin., *supra* note 16, at 3, 8.

²⁷ New York State Dep't. of Tax'n. & Fin., Sales and Use Tax Penalties, TB-ST-805, at 1 (2010).

which are subject to tax, and can calculate a total taxable amount on which to assess an ad valorem tax.²⁸ The retailer already records information on their transactions to track inventory, making it trivial to report taxable transactions.²⁹ And the retailer needs only to submit a single payment to remit on behalf of multiple consumers.³⁰ Another reason is that, from the perspective of the government, the cost of tax enforcement is lower when a single retailer facilitates tax payment.³¹ There is one point of contact, one set of records to audit and one party, if necessary, to levy sanctions against.³²

Another example of a tax where an “other party” to a transaction is required to collect and report is the individual income tax as it applies to wages and salary.³³ The statutory assignee is the individual worker who earns compensation in the context of an employment relationship and owes income tax.³⁴ However, the employer is required to assess and withhold that individual's expected federal income tax liability from each paycheck, and then remit it on a monthly or semi-monthly schedule.³⁵ In addition, the firm must file a Form W-2 for each employee at the end of the tax year, which informs the government of the total amount the employee was paid in that tax year (the

²⁸ N.Y. State Dep't. of Tax'n. & Fin., *Recordkeeping Requirements for Sales Tax Vendors*, Tax Bulletin ST-770 (Updated June 2, 2011)

²⁹ Kleven et. al., *supra* note 9, at 235.

³⁰ See N.Y. State Dep't. of Tax'n. & Fin., *Filing Requirements for Sales and Use Tax Returns*, Tax Bulletin ST-275 (updated Nov. 20, 2015) (explaining that a sales-and-use-tax return summarizes the vendor's gross sales, taxable sales, and sales taxes collected or required to be collected during the filing period); N.Y. State Dep't of Tax'n & Fin., *Filing a Final Sales Tax Return*, Tax Bulletin ST-265 (updated June 13, 2014) (explaining that an authorized business reports and remits the sales tax collected during each applicable sales-tax period).

³¹ Dhammika Dharmapala, Joel Slemrod & John D. Wilson, *Tax Policy and the Missing Middle: Optimal Tax Remittance with Firm-Level Administrative Costs*, 95 J. PUB. ECON. 1036, 1037–38 (2011).

³² See *id.* at 1038.

³³ See I.R.C. § 3402(a)(1) (requiring employers to deduct and withhold income tax from wages); § 6051(a) (requiring employers to report wages and taxes withheld).

³⁴ See *id.*; I.R.C. §§ 1, 61(a)(1).

³⁵ IRS, *About Form W-2, Wage and Tax Statement* (Jan. 29, 2024), <https://www.irs.gov/forms-pubs/about-form-w-2>.

firm must also furnish a copy of this report to the employee herself).³⁶

Voluntary tax compliance rates on transactions subject to both collection and reporting is generally high.³⁷ While there is considerable variation in state sales tax regimes, most estimates of average state sales tax compliance range from 70-90%.³⁸ In contrast, compliance rates for state use taxes, which apply sales tax rates to out of state purchases, are estimated to range from 8-15%.³⁹ Crucially, use taxes are self-assessed and remitted by the consumer (i.e., they are subject to neither collection nor reporting).⁴⁰ The discrepancy in collection rates is even more pronounced at the federal level: the IRS estimated that net underreporting of income from wages, tips, and salary was less than 1% for tax years 2014-16.⁴¹ In the same period, net underreporting on sole proprietor income, which is not subject to withholding and only minimal reporting, was estimated to be over 40%.⁴²

Information Reporting without Tax Remittance. Firms that are required to provide timely and specific information (but are not required to remit) are often referred to as third-party reporters.⁴³ Most third-party reporters are involved in financially processing the transaction, but do not have a tax incentive to underreport (in contrast to the other party to the transaction, who might benefit if

³⁶ *Id.*

³⁷ IRS, *The Tax Gap* (Oct. 23, 2023), <https://www.irs.gov/newsroom/the-taxgap#:~:text=The%20voluntary%20compliance%20rate%20is,compliance%20rate%20is%2083.7%20percent>; Joel Slemrod & Christian Gillitzer, *Insights from a Tax-Systems Perspective 15* (Univ. of Mich. Ross Sch. of Bus., Working Paper No. 1206, 2013).

³⁸ *Id.*

³⁹ United States Gov't Accountability Office, *States Could Gain Revenue from Expanded Authority, but Businesses Are Likely to Experience Compliance Costs*, GAO-18-114, at 5 available at <https://www.gao.gov/assets/gao-18-114.pdf>.

⁴⁰ *Id.*, at 5. Use taxes are the parallel tax that states levy on out of state purchases by their residents (until the 2018 Wayfair decision, almost all online purchases fell into this category).

⁴¹ IRS, *Federal Tax Compliance Research: Tax Gap Estimates for Tax Years 2014–2016*, Pub. No. 1415, at 20 (2022).

⁴² *Id.*

⁴³ IRS, *Third-Party Reporting Reminders*, FS-2006-24, at 1 (2006).

transactions went untaxed).⁴⁴ Joint liability is seldom imposed on third-party reporters, but they can be subject to fines if they do not provide the government with the required information in a timely manner (e.g., failure to file penalties).⁴⁵ Returning to the retail example, if the customer paid using a Visa credit or debit card, Visa is a third-party reporter required to file an annual 1099-K with the federal government on gross receipts processed through its system for most retailers.⁴⁶

Why would the government impose information reporting but not collection obligations? Generally, third-party reporters have recorded the information about a transaction for other purposes than firms required to collect and report.⁴⁷ For example, a brokerage firm must maintain records of its customers' short-term capital gains in the ordinary course of business.⁴⁸ However, unlike employers, the net effect of gains and losses fluctuate dramatically throughout the tax year, making accurate withholding challenging (wage income tax liability is more predictable).⁴⁹ Similarly, a firm that makes a payment to an independent contractor with an unknown number of other clients may not be well positioned to accurately withhold because they have insufficient information to predict the contractor's annual income.⁵⁰ However, they still must report this information as it is

⁴⁴ I.R.C. § 6721; Joshua D. Blank & Ari Glogower, *The Tax Information Gap at the Top*, 108 IOWA L. REV. 1597, 1615, 1632–1633 (2023).

⁴⁵ IRS, *Form 1099-K FAQs: Third party filers of Form 1099-K* (Mar. 19, 2024), <https://www.irs.gov/newsroom/form-1099-k-faqs-third-party-filers-of-form-1099-k>.

⁴⁶ These reporting requirements are often referred to as market maker or platform reporting rules and are relatively recent. For example, third party reporting requirements via the 1099-K were not implemented until 2012, *see* Notice 2011-89, 2011-46 I.R.B. 748, 748–749 (explaining that section 6050W was added by the Housing Assistance Tax Act of 2008 and that the first Forms 1099-K were filed in 2012 with respect to payments made in calendar year 2011).

⁴⁷ Kleven et. al., *supra* note 9, at 219–220.

⁴⁸ FINRA, *Cost Basis Basics—Here's What You Need to Know*, <https://www.finra.org/investors/insights/cost-basis-and-your-taxes>.

⁴⁹ Daniel McKeever & Kristian Rydqvist, *Tax-Loss Harvesting Under Uncertainty*, 140 J. BANKING & FIN. 106528, 1–2 (2022); James M. Poterba & Scott J. Weisbenner, *Capital Gains Tax Rules, Tax-Loss Trading, and Turn-of-the-Year Returns*, 56 J. FIN. 353, 353, 355–56 (2001).

⁵⁰ Kathleen DeLaney Thomas, *Taxing the Gig Economy*, 166 U. PA. L. REV. 1415, 1445–46 (2018).

a critical mechanism for ensuring the contractor accurately reports that income themselves.⁵¹

Information reporting has a long history within the U.S. federal income tax, with basic financial transaction reporting requirements introduced in 1917 preceding the 1942 establishment of firm withholding on wages.⁵² With the advent of electronic payment systems that create “paper” trails of transactions and tax software with the capacity to automate filing, the compliance costs of information reporting to modern businesses have declined significantly, making it an increasingly attractive tool for tax authorities.⁵³ For example, in 2012, the U.S. introduced Form 1099-K, which requires “market-makers” to report gross receipts processed for individuals receiving more than \$20,000 or 200 transactions.⁵⁴ This not only applied to traditional firms like Visa and Mastercard, but it also applied to digital platform firms like eBay and Uber.⁵⁵

The effect of information reporting on voluntary compliance varies significantly by context. Information reporting on financial transactions is estimated to reduce income underreporting by 4-8%.⁵⁶ However, the evidence on gross receipts reporting is more mixed. Public finance papers looking at the effect of Form 1099-K roll out have found that while information reporting does increase taxpayer reported gross income, it does not substantially increase

⁵¹ *Id.* at 1430.

⁵² IRS, *IRS History Timeline* (2019), <https://www.irs.gov/irs-history-timeline>.

⁵³ Anne Brockmeyer & Magaly Sáenz Somarriba, *Electronic Payment Technology and Tax Compliance: Evidence from Uruguay’s Financial Inclusion Reform*, 17 AM. ECON. J.: ECON. POL’Y 242, 247, 250–51 (2025); Siamand Hesami, Hatice Jenkins & Glenn P. Jenkins, *Digital Transformation of Tax Administration and Compliance: A Systematic Literature Review on E-Invoicing and Prefilled Returns*, 5 DIGIT. GOV’T: RSCH. & PRAC., art. 18, at 18:2, 18:7, 18:9 (2024).

⁵⁴ Kelly Phillips Erb, *New Credit Card Reporting Requirement Worries Some Taxpayers*, *Forbes* (Oct. 25, 2011), <https://www.forbes.com/sites/kellyphillipserb/2011/10/25/new-credit-card-reporting-requirement-worries-some-taxpayers/?sh=4f317857330b>.

⁵⁵ See Internal Revenue Service, *Understanding Your Form 1099-K* (explaining that Form 1099-K reporting applies to online marketplaces, auction sites, and car-sharing or ride-hailing platforms); eBay, *eBay and Form 1099-K*; Uber, *Tax Season Guide for Uber Drivers and Couriers*.

⁵⁶ IRS, *Estimation of the Underreporting Tax Gap for Tax Years 2014–2016: Methodology*, at 12 (2022).

reported income tax liability because many small businesses significantly inflated deductions.⁵⁷

Neither Information Reporting nor Tax Remittance. There are taxes in which firms play no part, either by design or because of logistical or political constraints. The government relies entirely on taxpayers to accurately assess, report and remit the relevant tax liability. One example, already mentioned above, are state use tax regimes. Use taxes typically apply when a consumer makes a purchase in a state other than the one in which they are a resident and that state has a lower sales tax rate than the consumer's home state. Consider a resident of Minnesota who drives across the western border to a shopping mall in North Dakota where they purchase a new kayak for \$1,000. At check out, they are charged ND's five percent state sales tax.⁵⁸ The sales tax rate in MN is higher, at 6.875 %.⁵⁹ Under MN's use tax law, the resident is required to remit the difference (5% vs. 6.875%, or \$18.75) to the MN Department of Revenue.⁶⁰

Historically, the federal income tax on cash tips is another prominent example of a tax that was subject to neither reporting nor remittance. The employer, it was argued, did not handle tips or record their value in the regular operation of their business. In the early 1970s, the Tax Court approved the IRS's use of aggregate estimation methods to reconstruct unreported tip income, but auditing individuals was costly and did not meaningfully improve

⁵⁷ Joel Slemrod, Brett Collins, Jeffrey L. Hoopes, Daniel Reck & Michael Sebastiani, *Does Credit-Card Information Reporting Improve Small-Business Tax Compliance?*, 149 J. PUB. ECON. 1 (2017); Bibek Adhikari, James Alm & Timothy F. Harris, *Small Business Tax Compliance Under Third-Party Reporting*, 203 J. PUB. ECON., art. 104514 (2021).

⁵⁸ See N.D. Cent. Code § 57-39.2-02.1 (imposing a five percent sales tax on the gross receipts of retailers). North Dakota does not exempt nonresidents from its sales tax; the tax attaches at the point of sale regardless of the buyer's domicile.

⁵⁹ Minn. Stat. § 297A.62, subdiv. 1 (imposing a general sales tax of 6.875 percent on gross receipts from retail sales).

⁶⁰ See Minn. Stat. § 297A.63 (imposing a use tax on tangible personal property stored, used, or consumed in Minnesota); *id.* § 297A.80 (allowing a credit against Minnesota use tax for sales or use tax legally imposed and paid to another state, to the extent of the rate imposed by that state).

system wide compliance, which was estimated to be around 8%.⁶¹ Congress ultimately responded with the Tax Equity and Fiscal Responsibility Act of 1982 which conscripted employers as information intermediaries.⁶²

Tax Remittance without Information Reporting. In certain circumstances, firms may choose to assist the government by remitting individual tax liabilities in the aggregate, without reporting information on either the underlying transactions or the identities of the specific individuals who incurred the liability.⁶³ While rare, these arrangements are not unprecedented, and may become more common as local tax authorities confront revenue loss from platform firms that grow rapidly and often exist in a legal gray zone in their early years.⁶⁴

The two most significant historical examples of remittance-only agreements are the Qualified Intermediary Program established in 2001 by the IRS⁶⁵ and the use of aggregate

⁶¹ See *McQuatters v. Comm'r*, 32 T.C.M. (CCH) 1122 (1973); see also *United States v. Fior D'Italia, Inc.*, 536 U.S. 238 (2002); U.S. Gen. Accounting Office, GAO/GGD-86-119, *Tax Administration: Tip Income Reporting Can Be Increased* 8–19 (1986) (discussing wide spread under reporting despite auditing).

⁶² Tax Equity and Fiscal Responsibility Act of 1982, Pub. L. No. 97-248, § 314, 96 Stat. 324; I.R.C. § 6053(a), (b), (c) (tip reporting requirements); I.R.C. § 6652(b) (penalty for failure to report tips). See also Joint Comm. on Tax'n, General Explanation of the Revenue Provisions of the Tax Equity and Fiscal Responsibility Act of 1982, at 199–200 (Dec. 31, 1982); Masood Chotani, *The IRS's Efforts to Improve Reporting of Tip Income*, 10 TAX DEV. J. 41, 42–46 (2020).

⁶³ See, e.g. *Voluntary Collection Agreement for Sonoma County Transient Occupancy Tax*, 1–2 (2016); Paris Martineau, *Inside Airbnb's "Guerrilla War" Against Local Governments*, Wired, <https://www.wired.com/story/inside-airbnbs-guerrilla-war-against-local-governments/> (Mar. 20, 2019).

⁶⁴ See Elizabeth Pollman & Jordan M. Barry, *Regulatory Entrepreneurship*, 90 S. CAL. L. REV. 383, 383–84, 398–401 (2017) (discussing scalable platform firms that operate in legal gray areas, seek to grow “too big to ban,” and attempt to alter state and local law); Shu-Yi Oei & Diane M. Ring, *Can Sharing Be Taxed?*, 93 WASH. U. L. REV. 989, 998, 1046 (2016).

⁶⁵ See U.S. Gov't Accountability Off., GAO-08-99, *Tax Compliance: Qualified Intermediary Program Provides Some Assurance That Taxes on Foreign Investors Are Withheld and Reported, but Can Be Improved*, at 9–10 (2007) (explaining that foreign financial institutions entering QI agreements

remittance agreements between Amazon and other large online retailers and U.S. states.⁶⁶

The Qualified Intermediary (QI) regime began in 2001, in response to concerns about tax evasion by Americans with foreign bank accounts.⁶⁷ A QI is any foreign intermediary that has entered into a qualified intermediary withholding agreement with the IRS.⁶⁸ The QI can certify treaty residence on behalf of its direct clients without revealing their identity. QIs are in overseas locations, and thus more likely to have personal contact with non-resident aliens.⁶⁹ Originally, QIs were allowed to hold US-persons accounts without disclosing their identity.⁷⁰ This aspect of the agreement was controversial, with the GAO finding billions of dollars flowing to unidentified recipients under the QI program, a gap plausibly attributable to the program's pooled non-disclosure reporting structure.⁷¹ In 2014, new legislation eliminated the non-disclosure pooling rule for US-persons accounts, while the foreign-persons pooling rule persists.⁷²

The VCAs between Amazon and other retailers obligated Amazon to collect sales tax on sales to a state's residents despite

may assume withholding responsibilities and report income and withholding on a pooled basis while preserving the anonymity of certain foreign account holders); N.Y. State Bar Ass'n. Tax Sec., *Report on Qualified Intermediary and Related Withholding and Information Reporting Legislation Proposed by the Administration*, Report No. 1189, at 10–11, 17 (Sept. 10, 2009); see also Press Release, U.S. Dep't. of Just., *Three Swiss Banks Reach Resolutions Under Justice Department's Swiss Bank Program* (June 9, 2015).

⁶⁶ See Yeliz Kaçamak, Tejaswi Velayudhan & Eleanor Wilking, *Retailer Remittance Matters: Evidence from Voluntary Collection Agreements*, 76 NAT'L TAX J. 239, 240, 242–44 (2023) (studying states' staggered adoption of voluntary collection agreements that committed large online retailers to remit sales taxes without corresponding information-reporting obligations). See also William F. Fox, Enda Patrick Hargaden & LeAnn Luna, *Statutory Incidence and Sales Tax Compliance: Evidence from Wayfair*, 213 J. PUB. ECON. 104716, at 2 (2022) (describing pre-Wayfair VCAs between individual states and large online retailers such as Amazon).

⁶⁷ GAO, *supra* note 65, at 9.

⁶⁸ *Id.* at 11.

⁶⁹ *Id.* at 9.

⁷⁰ N.Y. State Bar Ass'n. Tax Sec., *supra* note 65, at 10.

⁷¹ GAO, *supra* note 65, at 11.

⁷² *Id.*

the absence of nexus and then remit this liability in aggregate.⁷³ Originally, some states enacted laws that directly targeted online retailers like Amazon to force them to remit taxes. For example, Rhode Island first enacted a law targeting retailers, known as the “Amazon Tax,” in 2009.⁷⁴ At that time, Amazon responded to by immediately cutting ties with all Rhode Island businesses. However, Amazon changed course and began to enter into VCAs in roughly a dozen states in 2017.⁷⁵ Some speculated that this was a strategic move, although Amazon and states were secretive as to why it began to enter into these agreements.⁷⁶

While the 2018 *South Dakota v. Wayfair* decision, in which the Supreme Court relaxed nexus requirements, made many of these agreements obsolete, their widespread adoption raised many questions.⁷⁷ Both of the examples above highlight the complexity of allowing a third-party to remit, but not report, taxable transactions, and raise the following questions: Why would a firm “volunteer” to assist the government in collecting tax for which it is not liable on transactions it cannot be compelled to report? When should the tax authority accept such an offer? Does doing so significantly increase tax compliance? If so, at what cost?

⁷³ Sales Tax Institute, *Amazon to Begin Collecting Wyoming Sales Tax on March 1, 2017* (Jan. 31, 2017), <https://www.salestaxinstitute.com/resources/amazon-begin-collecting-wyoming-sales-tax-march-1-2017>. See also Kacamak et. al, *supra* note 66, at 240-2.

⁷⁴ See R.I. Gen. Laws § 44-18-15 (2009) (establishing click-through nexus for out-of-state retailers with in-state affiliates generating more than \$5,000 in referral sales); see also Joseph Henchman, “Amazon Tax” Laws Signal Business Unfriendliness and Will Worsen Short-Term Budget Problems, Tax Found. (Feb. 3, 2010), <https://taxfoundation.org/research/all/state/amazon-tax-laws-signal-business-unfriendliness-and-will-worsen-short-term-budget-problems/>

⁷⁵ Henchman, *supra* note 74.

⁷⁶ Michelle Quinn, Amazon Cuts Sales Tax Deals with States Not Feds (May 1, 2012), <https://www.politico.com/story/2012/05/amazon-cuts-sales-tax-deals-with-states-not-feds-075803>; Kelly Phillips Erb, Tax-Free No More: Amazon To Begin Collecting Sales Tax Nationwide on April 1, Forbes (Mar. 27, 2017), <https://www.forbes.com/sites/kellyphillips/2017/03/27/tax-free-no-more-amazon-to-begin-collecting-sales-tax-nationwide-on-april-1/> (“A request for comment from Amazon about the change was not immediately returned.”); Jessica Melugin, Utah Keeping Amazon Sales Tax Deal Secret (Jan. 6, 2017), <https://cei.org/blog/utah-keeping-amazon-sales-tax-deal-secret/Jessica>.

⁷⁷ *S. Dakota v. Wayfair, Inc.*, 585 U.S. 162 (2018).

II. A STUDY OF AIRBNB VOLUNTARY COLLECTION AGREEMENTS

This section presents an original case study of voluntary collection agreements signed by Airbnb. The two examples above of previous VCAs indicate that the agreements can be made in a variety of different sectors, however, both are now mostly moot, at least when it comes to US persons.⁷⁸ The VCAs in the hotel industry are critical to study as there are hundreds of individual agreements addressing the economic, political, and legal ramifications of large-scale voluntary tax collection replicated across hundreds of individually negotiated agreements, rather than a single federal framework or a handful of state deals.

After explaining how these agreements came to be, and what terms they contained, I empirically assess their direct effect on tax revenue collection—as well as certain unintended consequences for other regulatory objectives.

A. Hotel Taxes

Hotel taxes, also referred to as “occupancy taxes” or “short-term occupancy taxes,” are levied by many U.S. cities on hotel stays.⁷⁹ They are structured and effectively operate as a special retail tax, imposed in addition to the local sales tax, and calculated as a percentage of the purchase price.⁸⁰ Each city defines what constitutes a short-term stay, but most restrict it to stays shorter than six weeks.⁸¹ Cities also set the rate of hotel taxes, which can range from 3% (Phoenix, AZ)⁸² to nearly 17% (Washington, DC)⁸³.

⁷⁸ See *supra* Section II.B.

⁷⁹ Nat’l. Conf. of State Legis., *Specific Statewide Taxes on Lodging - By State* (Oct. 20, 2020), <https://www.ncsl.org/elections-and-campaigns/state-lodging-taxes>.

⁸⁰ In this regard, hotel taxes are like “sin taxes” levied by local government on products such as alcohol and cigarettes.

⁸¹ Avalara, *Lodging Tax Requirements by State* (Sept. 30, 2021), <https://www.avalara.com/mylodgetax/en/resources/state-lodging-tax-requirements.html>.

⁸² Ariz. Off. of Tourism, *Transient Lodging Tax Rates: Arizona Communities* (2021), <https://tourism.az.gov/wp-content/uploads/2021/01/Bed-Tax-Rates-Alpha-January-2021.pdf>.

⁸³ Martin Austermuhle, *D.C. Will Increase its Hotel Tax to Pay for a Marketing Campaign to Entice Tourists to Visit*, DCist (Dec. 22, 2022),

These taxes are nominally assessed on the consumer (the hotel guest) but are collected by the hotel and sent monthly to the city's revenue department.⁸⁴ Until the entry of digital platform firms, this system was straightforward; it was considered a relatively stable source of revenue for annual budgeting and enjoyed the support of local policymakers, likely because the tax was levied on tourists rather than city voters.⁸⁵

Many cities, particularly those with large tourism industries, rely on hotel tax revenues.⁸⁶ These tax revenues can form a significant part of the cities' overall sales tax budgets.⁸⁷

<https://dcist.com/story/22/12/22/dc-hotel-tax-increase-tourism-marketing-20-million/>.

⁸⁴ See, e.g., Tex. Tax Code Ann. § 156.051 (West 2023) (imposing the hotel occupancy tax on the person paying for the room); *id.* § 156.052 (requiring the hotel operator to collect the tax and remit it to the Comptroller). The hotel occupancy tax is nominally a tax on the consumer—the hotel guest—but the operator acts as a fiduciary collection agent for the taxing authority. See Roberta A. Kaplan & Michael L. Nadler, *Airbnb: A Case Study in Occupancy Regulation and Taxation*, 82 U. CHI. L. REV. ONLINE 103, 108 (2015).

⁸⁵ See GOV'T FIN. OFFICERS ASS'N, *Tourism and Lodging Taxes* (Mar. 2020), <https://www.gfoa.org/materials/tourism-and-lodging-taxes> (describing tourism and lodging taxes as “a main source of revenue for local government” in resort areas). Hotel taxes enjoy particular political support because their economic incidence is largely exported to nonresident visitors who cannot vote in local elections. See Edwin T. Fujii, Mohammed Khaled & James Mak, *The Exportability of Hotel Occupancy and Other Tourist Taxes*, 38 NAT'L TAX J. 169, 175–76 (1985) (finding that a hotel room tax is “readily, though not fully, shifted” to visitors); see also Rita Jefferson, *Travelers' Checks: How to Tax Tourists in States and Localities*, INST. ON TAX'N & ECON. POL'Y (Mar. 30, 2026), <https://itep.org/how-to-tax-tourists-in-states-and-localities/> (noting that hotel taxes are “primarily paid by non-residents”).

⁸⁶ See GOV'T FIN. OFFICERS ASS'N, *supra* note 85; Thomas Hazinski, *2025 HVS Lodging Tax Report – USA*, HVS (2025), <https://www.hvs.com/article/10299-2025-lodging-tax-report-usa> (finding that more than 120 of the 150 largest U.S. cities impose a dedicated lodging tax); see also Jefferson, *supra* note 85 (noting that hotel tax revenue funds infrastructure, environmental protection, and public safety in tourism-dependent cities).

⁸⁷ See Thomas Hazinski, *2025 HVS Lodging Tax Report – USA*, HVS (2025), <https://www.hvs.com/article/10299-2025-lodging-tax-report-usa> (reporting approximately \$4.19 billion in lodging tax revenue across 25 major U.S. markets and noting that lodging taxes are “a significant source of funding for economic development initiatives” in tourist destinations); see, e.g., *Palo Alto City Leaders Worried Airbnb Cutting into Hotel Tax Revenue*, CBS NEWS

Hotel taxes are used to fund, among other things, infrastructure that directly supports the tourism industry: operating museums, maintaining beaches, and so on.⁸⁸ They are also used to build or repair infrastructure used by both tourists and residents alike, such as transportation, to limit the crowding out of public goods for residents.⁸⁹

Enter Airbnb. Airbnb is the largest firm facilitating short-term, peer-to-peer residential space rentals through an online platform.⁹⁰ The firm was started in 2008 as an online marketplace to connect couch surfers and initially represented a negligible share of the aggregate hotel market.⁹¹ However, Airbnb has experienced remarkable growth in the last decade and a half, with listings expanding exponentially in popular tourism destinations around the globe.⁹²

As Airbnb grew, so did city administrators' frustration with Airbnb hosts' avoidance of hotel taxes.⁹³ In cities with significant

(Aug. 14, 2014), <https://www.cbsnews.com/sanfrancisco/news/palo-alto-city-leaders-worried-airbnb-cutting-into-hotel-tax-revenue> (reporting that Palo Alto's transient occupancy tax makes up approximately seven percent of the city's budget).

⁸⁸ See Hazinski, *supra* note 87.

⁸⁹ *Id.*

⁹⁰ As of 2024, Airbnb held approximately forty-four percent of the global short-term rental market, more than double its nearest competitor. See Seth Borko, *Short-Term Rentals: Airbnb's Dominance and Booking's Gains in 1 Chart*, SKIFT RSCH. (Mar. 14, 2025), <https://skift.com/2025/03/14/short-term-rentals-airbnbs-dominance-and-bookings-gains-in-1-chart/>; see also Vivienne Walt, *Paris Is Ground Zero for Europe's Backlash Against Illegal Airbnbs*, FORTUNE (Mar. 27, 2026), <https://fortune.com/2026/03/27/airbnb-paris-illegal-listing-housing-market/> (noting that Airbnb has approximately nine million listings globally).

⁹¹ Airbnb, *About Us*, <https://news.airbnb.com/about-us/> (last visited Apr. 29, 2026).

⁹² Vivienne Walt, *Paris Is Ground Zero for Europe's Backlash Against Illegal Airbnbs*, Fortune (Mar. 27, 2026), <https://fortune.com/2026/03/27/airbnb-paris-illegal-listing-housing-market/> (claiming Paris has an estimated 75,000 active Airbnb listings, the most of any city in the world.)

⁹³ See Kaplan & Nadler, *supra* note 84, at 111–13 (documenting cities' early frustration with widespread noncompliance among Airbnb hosts); see also *Palo Alto City Leaders Worried Airbnb Cutting into Hotel Tax Revenue*, CBS NEWS (Aug. 14, 2014), <https://www.cbsnews.com/sanfrancisco/news/palo-alto-city-leaders-worried-airbnb-cutting-into-hotel-tax-revenue> (reporting that only

levels of tourism, the estimated loss of hotel tax revenue was substantial.⁹⁴ But why couldn't cities easily collect hotel tax on transactions that clearly fell within their short term occupancy definitions? In Section II.B, *infra*, I explore why cities struggled to enforce them against Airbnb hosts or the platform itself.

B. *Challenges in Taxing Airbnb*

Despite the numerous problems posed by regulatory arbitrage, Airbnb and similar platforms have proven remarkably difficult for cities to tax and regulate. Some of this is an old story of innovation running ahead of regulatory capacity. But there are also structural features of platforms that pose truly novel regulatory challenges. In this section, I identify inter-related economic, legal, and political characteristics of Airbnb that make tax and regulatory enforcement difficult. While identified in the specific context of Airbnb rentals, the concerns generalize to most other platform firms.

1. Economic

There are several features of Airbnb's business model that present distinct challenges, particularly for regulators accustomed to the traditional hotel industry.

Points of collection and economies of enforcement. A key finding in the public finance literature is that there is less tax evasion the higher up the supply chain collection occurs.⁹⁵ The mechanisms driving this phenomenon are applicable to Airbnb. First, because enforcement resources are scarce, they can be used far more efficiently to collect tax revenues from a few large firms than many small ones.⁹⁶ Second, larger firms are more likely to

two Airbnb hosts in Palo Alto had registered to remit occupancy taxes despite the tax comprising seven percent of the city's budget); Kevin Kelleher, *Airbnb Offers To Pay Hotel Taxes in NY, Hotel Lobby Says 'No Thanks,'* TECHCRUNCH (Sept. 16, 2014), <https://techcrunch.com/?p=989530>.

⁹⁴ *All Eyes on the Sharing Economy*, Economist (Mar. 9, 2013),

<http://www.economist.com/news/technology-quarterly/21572914-collaborative-consumption-technology-makes-it-easier-people-rent-items>.

⁹⁵ Kopczuk, Marion, Muehlegger & Slemrod, *supra* note 15, at 260–65.

⁹⁶ *Id.* at 266–8.

create a paper trail for transactions than smaller firms, easing the task of the tax collector.⁹⁷ Finally, larger firms are typically more risk averse about outright evasion than smaller firms, usually because they have more to lose.⁹⁸ In Airbnb's case, because the hotel tax is levied at the host level, prior to VCAs, cities had to rely on thousands of individual hosts being both aware of, and willing to fulfill, their obligation to collect hotel taxes.⁹⁹ This represented exponentially more points of collection for hotel taxes than cities were used to and made cost effective auditing impossible.¹⁰⁰ For example, in Chicago's central loop district, there are approximately 140 registered hotels but more than two thousand Airbnb listings.¹⁰¹ Though Airbnb's position was that hosts were supposed to "factor in local taxes" in setting their initial price and remit out

⁹⁷ See Henrik Jacobsen Kleven, Claus Thustrup Kreiner & Emmanuel Saez, *Why Can Modern Governments Tax So Much? An Agency Model of Firms as Fiscal Intermediaries*, 83 *ECONOMICA* 219, 221–24 (2016) (modeling how accurate business records in larger firms make collusive tax evasion unsustainable because any employee can reveal the true records to the government).

⁹⁸ Joel Slemrod, *Tax Compliance and Enforcement*, 57 *J. ECON. LITERATURE* 904, 921–23 (2019) (reviewing evidence that larger firms face higher expected costs from detected evasion, including reputational damage and greater audit probability, inducing higher compliance); see also Kleven, Kreiner & Saez, *supra* note 97, at 225–26 (noting that the proportion of informal workers diminishes with firm size).

⁹⁹ See Kaplan & Nadler, *supra* note 84, at 108–10 (describing the shift from a small number of hotel operators to thousands of dispersed individual hosts, each independently responsible for tax remittance); see also Liran Einav, Chiara Farronato & Jonathan Levin, *Peer-to-Peer Markets*, 8 *ANN. REV. ECON.* 615, 630–32 (2016) (discussing the regulatory challenges posed by the decentralized structure of platform markets).

¹⁰⁰ See Kopczuk, Marion, Muehlegger & Slemrod, *supra* note 15, at 254–55 (demonstrating that tax compliance declines as collection is pushed to more numerous, smaller entities); Kaplan & Nadler, *supra* note 84, at 110–11 (noting that the volume of individual hosts made traditional enforcement methods impracticable for municipal tax authorities).

¹⁰¹ See Choose Chicago, Chicago Hotel Supply, <https://www.choosechicago.com/press-media/toolkit/tourism-research-and-statistics/chicago-hotel-supply/> (last visited Aug. 12, 2026) (listing 151 hotel properties in Chicago's central business district with 45,685 rooms); Chicago Airbnb Data, AirDNA (2024), <https://www.airdna.co/vacation-rental-data/app/us/illinois/chicago/overview> (reporting approximately 2,000 active short-term rental listings in Chicago's "Central Loop" region.)

of their “share” of the transaction, very few hosts did.¹⁰² It did not help that city collection systems were designed for large hotel accountants, rather than small businesses. Many cities had no dedicated digital portal for short-term rental hosts to remit hotel taxes, instead requiring hosts to independently register with the local tax authority, file periodic paper returns, and mail payments to the appropriate city agency.¹⁰³ As a result, compliance was minimal—in Palo Alto, for example, only two Airbnb hosts had registered to pay the city's transient occupancy tax as of 2013, and only one had actually remitted payment.¹⁰⁴

Some cities experimented with host registration systems, but without access to host contact information, it was challenging to communicate this to hosts, much less assess how many hosts complied with registration.¹⁰⁵ I discuss these initiatives in further detail in Section II.D, *infra*.

Limited physical capital requirements. Another difference between Airbnb and traditional hotels concerns the upfront capital required for operation. Hotels require large capital investments in

¹⁰² See Airbnb, Terms of Service § Host Responsibilities, <https://www.airbnb.com/help/article/2908> (last visited Aug. 12, 2026) (stating that hosts are “solely responsible for determining [their] obligations to report, collect, remit or include in [their] Listing Fees any applicable . . . occupancy tax”); Kaplan & Nadler, *supra* note 84, at 111–13 (documenting widespread noncompliance among Airbnb hosts with local hotel tax obligations). See also Anders Munk-Nielsen & Christoffer Ruud Bjerregaard, *The Hidden Costs of the Platform Economy: Tax Dishonesty by Airbnb Hosts* (CBS Research Portal, 2024) (finding approximately 658 million DKK in unreported Airbnb income in the Copenhagen area during 2017–2019).

¹⁰³ See Gov't Fin. Officers Ass'n, *supra* note 85 (describing the administrative challenges municipalities face in collecting lodging taxes from short-term rental hosts).

¹⁰⁴ See also Gennady Sheyner, *Airbnb Renters May Owe Palo Alto Taxes*, Palo Alto Online (June 9, 2013), <https://www.paloaltoonline.com/news/2013/06/09/airbnb-renters-may-owe-palo-alto-taxes-> (reporting that only two hosts had registered to pay the city's twelve-percent transient occupancy tax, and only one had actually paid).

¹⁰⁵ See Sarah Warren, *The Failure to Regulate Airbnb*, 23 CONTEXTS 32, 33–35 (2024) (describing how Airbnb's “data hoarding”—withholding information about hosts, listings, and pricing—made it “nearly impossible for cities to monitor Airbnb's activity or write clear regulations, let alone enforce them”); see also Murray Cox, *Inside Airbnb*, <https://insideairbnb.com/> (last visited Aug. 12, 2026) (independent watchdog project that scrapes Airbnb data and shares it with approximately thirty cities for enforcement).

the construction of new property or the renovation of an existing space, and require city permits at almost every stage.¹⁰⁶ Even without permitting, property construction is almost by definition highly visible: standing up a new hotel in the center of a city is a multi-year process that would be almost impossible to hide.¹⁰⁷

In contrast, Airbnb leverages the existing capital (i.e., properties) of individual owners. The conversion of an individual owner-occupied unit into a full-time, short-term rental takes days or months, not years. In addition, this process largely occurs without the public indicators that accompany hotel construction. Because the barriers to adding new property listings to Airbnb are so low, it can also be difficult for regulators to accurately estimate the size or growth of the market in real time.¹⁰⁸ In effect, Airbnb's business model allowed for a "ask forgiveness" model in what had always been an "ask permission" industry.

Exponential Growth. A related but distinct feature of Airbnb relative to traditional hotels is the nearly exponential growth rate in Airbnb listings that occurred in a relatively short period of time.¹⁰⁹ This phenomenon meant that many cities, accustomed to several years of slow Airbnb growth upon its arrival, were blindsided when the platform's popularity suddenly exploded,

¹⁰⁶ See HVS, U.S. HOTEL DEVELOPMENT COST SURVEY 2023 (reporting median per-room development costs ranging from \$148,000 for economy hotels to \$850,000 for luxury properties); see also Hotel Construction & Development Guide for 2026, PROPERTYBUILD (describing the multi-stage permitting process including building permits, fire department approvals, health department licenses, and certificate of occupancy).

¹⁰⁷ See Tyler Cauble, *How to Build a Hotel: The Development Timeline Nobody Talks About*, THE CAUBLE GRP. (2024) (estimating twelve to twenty-four months for hotel construction plus six to twelve months for pre-construction permitting and design).

¹⁰⁸ See Warren, *supra* note 105, at 37 (describing how Portland officials were left "back at the starting point, still just trying to understand the scope and scale of their city's short-term rental market"); see also Einav, Farronato & Levin, *supra* note 99, at 630–32 (discussing the regulatory challenges posed by the informality and rapid scalability of peer-to-peer platforms).

¹⁰⁹ See Airbnb Statistics, IPROPERTYMANAGEMENT (2026), <https://ipropertymanagement.com/research/airbnb-statistics> (reporting that listings grew from approximately 300,000 in 2013 to one million in 2015, and subsequently doubled roughly every one to two years through 2019).

leaving regulators scrambling to respond to a market that had already reached significant scale.¹¹⁰

Low barriers to entry for platforms and hosts. The last key economic feature of Airbnb and similar platform firms concerns their creation. Platforms themselves are relatively easy to stand up; they do not require a large physical capital investment, as a traditional hotel does.¹¹¹ The start-up cost of a new platform is the creation of a website and payment system.¹¹² As a result of these relatively low entry costs, if one platform is regulated or taxed, then a nearly identical one could quickly enter and take its place, leading to a regulatory game of ‘whack-a-mole.’ A side-by-side comparison of the Airbnb and VRBO interfaces suggests this inherent substitutability.¹¹³ Similarly, host profiles can be recreated on a new platform in minutes, and many hosts already list on multiple platforms.¹¹⁴

2. Legal

¹¹⁰ See Warren, *supra* note 102, at 32 (observing that “[t]he rapid global explosion in short-term rentals has left cities, armed only with policies related to hotels and permanent housing, scrambling”); see also Lobel, Only Lobel, *The Law of the Platform*, 101 MINN. L. REV. 87, 127–30 (2016) (describing how platform firms outpace regulatory capacity).

¹¹¹ See Lobel, *supra* note 110, at 127–30 (identifying “reduced barriers to entry” as a defining characteristic of platform firms and documenting how platforms leverage existing assets rather than making capital investments).

¹¹² See Einav, Farronato & Levin, *supra* note 99, at 618–20 (noting that peer-to-peer platforms’ primary costs are in software development and payment infrastructure, with value driven by network effects rather than physical capital).

¹¹³ Note that the counterargument to this assertion is the importance of network effects—if all customers want to book on Airbnb, demand for that specific platform may be sticky. While acknowledging this, it seems that in the long run, consumers will likely respond to substantial price differences. Further, the real effect of platform substitutability may be in the inhibition of regulators, who are reluctant to invest in regulating a single firm when another could take its place.

¹¹⁴ AirDNA, *Where to List Short-Term Rentals: Airbnb vs. Vrbo vs. Booking.com* (2024), <https://www.airdna.co/blog/where-to-list-short-term-rentals> (reporting that professional hosts commonly list on multiple platforms simultaneously to maximize occupancy); see Lobel, *supra* note 110, at 130–32 (discussing low switching costs and multi-homing in platform markets).

In addition to the inherent difficulty of regulating thousands of new rental spaces without location information in a short amount of time, cities also faced significant legal barriers. Airbnb used the novelty of its business model vis-à-vis existing hotel regulation to its advantage, first in disputing the applicability of STR regulations and then by aggressively pushing back on various attempts by local government to limit or exercise control over listings. This latter strategy bore much in common with Amazon's scorched earth legal battles over state sales taxation authority pre-Wayfair.

Characterization of Transactions. First, Airbnb took the position that its hosts were not subject to hotel tax under existing city regulations because host-guest transactions were “not commercial” in nature.¹¹⁵ This position had almost no basis in existing regulations, and quickly became politically untenable as Airbnb's market share increased.

Legal Uncertainty over City Regulatory Powers. The real legal issue quickly became to what extent Airbnb could be compelled to assist cities in collecting taxes or enforcing regulation. While New York was successful in obtaining basic retrospective listing data in *Airbnb, Inc. v. Schneiderman*,¹¹⁶ cities struggled to obtain useable information (e.g. by maintaining a database of hosts, listings, and bookings that the city could access in real time). And even when local governments prevailed in court, the litigation process could span years. In New York, Airbnb sued to block a city ordinance requiring it to turn over more detailed information on listings in 2019—the law will not take effect until mid 2024.¹¹⁷

¹¹⁵ Brian Chesky, *Who We Are, What We Believe, and Where We Are Headed*, Airbnb Blog (Oct. 3, 2013) (arguing that Airbnb hosts “are not hotels” and urging that peer-to-peer rentals should not be taxed like commercial lodging).

¹¹⁶ *Airbnb, Inc. v. Schneiderman*, 989 N.Y.S.2d 786 (2014). The court ruled that the attorney general was allowed to subpoena Airbnb to investigate taxes paid by hosts. It also appears to be a pretty low standard; the AG just needs to show that the request has a “reasonable relation” to the subject matter.

¹¹⁷ Alexander Lycoyannis, *NY Court: Short-Term Rental Services Must Register Dwellings to Comply with Local Law* (Aug. 17, 2023),

Cities were even less successful in persuading courts that Airbnb should be required to directly collect taxes on behalf of hosts at the point of transaction. In *Gannon v. Airbnb, Inc.*,¹¹⁸ a judge found that an attempt by Florida to make Airbnb (and other platforms) directly collect taxes was impermissible because “companies were not in the business of renting, leasing, or letting transient accommodations,” and were therefore not responsible for collecting the hotel tax.¹¹⁹

These efforts were taking place in parallel with litigation on when *state* governments have the authority to compel sales tax collection from out of state online retailers. While the Supreme Court eventually affirmed state powers to tax online purchases (*Wayfair vs. South Dakota*), extending the same logic to city hotel tax enforcement has been rebuffed by multiple courts.¹²⁰ While appearing inconsistent on the surface, Airbnb has successfully argued that cities are so varied in their regulations on STR, it would be ludicrous to require a corporation to separately remit to thousands of municipalities (i.e. complying with fifty state sales tax regulations is a reasonable burden to place on retailers).

3. Political

There were also political factors that stymied municipal regulators and tax authorities.

Limited Municipal Resources for Enforcement. Even where municipalities possessed the legal authority to collect transient occupancy taxes from short-term rental hosts, budgetary constraints severely hampered their practical capacity to do so. Municipal tax enforcement offices are staffed and resourced to address dominant revenue sources—property taxes, sales taxes, and local income taxes—against a manageable number of identifiable, registered taxpayers. The emergence of Airbnb disrupted this model by multiplying the universe of potentially

<https://www.hklaw.com/en/insights/publications/2023/08/ny-court-short-term-rental-services-must>.

¹¹⁸ *Gannon v. Airbnb, Inc.*, 295 So. 3d 779 (Fla. Dist. Ct. App. 2020).

¹¹⁹ There is also a discussion of whether Airbnb could be considered a “dealer,” which the judge also answered in the negative. A few months later, Florida state passed a Marketplace Facilitator Law that expressly excluded property rentals from liability.

¹²⁰ *Gannon v. Airbnb, Inc.*, 295 So. 3d 779 (Fla. Dist. Ct. App. 2020).

taxable entities from dozens of licensed hotel operators to thousands of anonymous private hosts, without any corresponding expansion of enforcement capacity.

Even holding fixed the issue of resources, enforcement agencies operating under finite budgets typically allocate investigative resources to their most productive targets. A single non-compliant hotel might represent tens or hundreds of thousands of dollars in annual tax liability; an individual Airbnb host renting a spare bedroom might represent a few hundred.

Rapid Creation of Sympathetic Constituencies. Airbnb's growth also generated a concentrated constituency of property owners with a direct financial stake in resisting municipal oversight. Many homeowners and apartment dwellers had incorporated rental income into their household finances and were likely to mobilize if that income was threatened.

Airbnb proved skilled at activating this constituency. When San Francisco—the company's home city—placed Proposition F on the November 2015 ballot, a measure that would have imposed significant restrictions on short-term rentals, Airbnb organized its host community into a grassroots opposition campaign while simultaneously spending an estimated eight million dollars on advertising against the measure.¹²¹ Hosts testified at city hearings, canvassed precincts, and flooded elected officials with constituent communications.¹²² The measure failed.¹²³

The political calculus for city officials was further complicated by widespread public antipathy toward the hotel industry. Hotels were broadly perceived as expensive, impersonal, and representative of entrenched corporate interests, while Airbnb successfully positioned itself—and its hosts—as participants in a

¹²¹ *Airbnb Spends More Than \$8 Million in San Francisco Fight*, *Time* (Nov. 3, 2015), <https://time.com/4096042/airbnb-san-francisco-proposition-f-campaign-rentals/>; City of San Francisco Initiative to Restrict Short-Term Rentals, Proposition F (November 2015), Ballotpedia, [https://ballotpedia.org/City_of_San_Francisco_Initiative_to_Restrict_Short-Term_Rentals,_Proposition_F_\(November_2015\)](https://ballotpedia.org/City_of_San_Francisco_Initiative_to_Restrict_Short-Term_Rentals,_Proposition_F_(November_2015)) (noting Airbnb contributed \$8,377,784—over 90% of the opposition's total war chest).

¹²² *Id.*

¹²³ *Voters Reject Anti-Airbnb Measure in San Francisco*, *CNBC* (Nov. 4, 2015), <https://www.cnn.com/2015/11/04/voters-reject-anti-airbnb-measure-in-san-francisco.html>.

democratizing "sharing economy" that empowered ordinary homeowners.¹²⁴ Municipal officials who aligned themselves with hotel industry complaints presumably risked appearing to subordinate constituent economic interests to those of a wealthy and unpopular incumbent industry.

Differences between local and state objectives. Airbnb also fought to override or pre-empt municipal short term rental restrictions by petitioning state legislators to intervene. For example, both Austin and Nashville attempted to pass legislation regulating short-term rental properties. In both cases, Airbnb waged aggressive, well-funded lobbying campaigns and successfully persuaded state lawmakers in Texas and Tennessee to pass legislation heavily restricting local governments' powers to regulate short-term rentals.¹²⁵

C. *Agreements Background & Terms*

Initially, Airbnb's position was that its rentals were not subject to hotel taxes because transactions were "peer-to-peer" rather than commercial in nature.¹²⁶ In May 2014, the company officially retracted this view and announced that it believed its hosts were responsible for paying hotel taxes to local governments. It also amended its "Terms of Service" agreement to inform hosts of their obligation to research and comply with applicable local taxes and

¹²⁴ See, e.g., Arun Sundararajan, *THE SHARING ECONOMY* (2016); Orly Lobel, *The Law of the Platform*, 101 MINN. L. REV. 87 (2016)

¹²⁵ Phil Williams, *REVEALED: Airbnb Legislation Shows How Capitol Hill Culture Affects Tennessee Communities* (Jan. 10, 2024, 11:16 AM), <https://www.newschannel5.com/news/newschannel-5-investigates/revealed-airbnb-legislation-shows-how-capitol-hill-culture-affects-tennessee-communities>; Mariana Alfaro, *Texas Proposal Would Keep Cities From Restricting Short-term Home Rentals* (Feb. 27, 2017, 12:00 AM), <https://www.texastribune.org/2017/02/25/bill-would-overrule-local-legislation-over-short-term-rentals/>.

¹²⁶ Shu-Yi Oei & Diane M. Ring, *Can Sharing Be Taxed?*, 93 WASH. U.L. REV. 989, 1024 (2016).

regulations.¹²⁷ Predictably, few hosts complied, and Airbnb began to negotiate.

On June 28, 2014, Airbnb announced that it had reached an agreement with the city of Portland, OR to collect an 11.5% hotel tax on all reservations booked on its site, and to remit these taxes to the city at the end of each quarter.¹²⁸ Crucially, the agreement explicitly prohibited Portland from requiring Airbnb to disclose information related to taxable transaction that could individually identify hosts.¹²⁹ As part of the exchange, the Portland City Council agreed to pass a code revision that would legalize short-term home rentals if residents obtained a \$180 permit and installed fire alarms.

Between August 2014 and August 2016, similar agreements to collect and remit hotel taxes were signed with San Francisco, CA (14.5%), San Jose, CA (10%), Chicago, IL (4.5%), Washington, DC (16.75%), Philadelphia, PA (8.5%), Durham, NC (6%), San Diego, CA (10.5%), and Phoenix, AZ (3%), as well as several smaller municipalities.¹³⁰ Typically, an agreement was announced two weeks before the date when Airbnb began collecting taxes on all bookings in that jurisdiction. Airbnb notified affected hosts via

¹²⁷ Beginning May 1, 2014, Airbnb's Terms of Service includes the following paragraph:

YOU AS A HOST UNDERSTAND AND AGREE THAT YOU ARE SOLELY RESPONSIBLE FOR DETERMINING (I) YOUR APPLICABLE TAX REPORTING REQUIREMENTS, AND (II) THE TAXES THAT SHOULD BE INCLUDED, AND FOR INCLUDING TAXES TO BE COLLECTED OR OBLIGATIONS RELATING TO APPLICABLE TAXES IN LISTINGS. YOU ARE ALSO SOLELY RESPONSIBLE FOR REMITTING TO THE RELEVANT AUTHORITY ANY TAXES INCLUDED OR RECEIVED BY YOU. AIRBNB CANNOT AND DOES NOT OFFER TAX-RELATED ADVICE TO ANY MEMBERS.

¹²⁸ *Transient Lodgings Filing and Payment Information*, <https://www.portland.gov/revenue/transient-lodgings-tax> (last visited Apr. 29, 2026).

¹²⁹ Elliot Njus, *Airbnb, acting as Portland's lodging tax collector, won't hand over users' names or addresses*, *Oregonian* (July 21, 2014), https://www.oregonlive.com/front-porch/2014/07/airbnb_acting_as_portlands_lod.html.

¹³⁰ See Wash. Post, *Airbnb is about to start collecting hotel taxes in more major cities including Washington*, <https://www.washingtonpost.com/news/wonk/wp/2015/01/29/airbnb-is-about-to-start-collecting-hotel-taxes-in-more-major-cities-including-washington/>; see also Martineau, *supra* note 63.

email shortly after the announcement. To date, Airbnb has signed VCAs with over 275 jurisdictions across the U.S.¹³¹

Understanding why Airbnb signed these agreements requires appreciating the regulatory threats the company faced and the strategic advantages VCAs offered in neutralizing them. By 2014, Airbnb confronted an increasingly hostile regulatory environment: New York had successfully subpoenaed host data,¹³² San Francisco was considering ballot measures to restrict short-term rentals,¹³³ and multiple cities were exploring whether existing hotel regulations could be applied to the platform directly. VCAs offered Airbnb a way to defuse these threats on favorable terms. By voluntarily collecting taxes, Airbnb could position itself as a cooperative corporate citizen—willing to contribute to city coffers—while securing contractual protections against the regulatory actions it feared most. The agreements shielded host-identifying data from municipal authorities, foreclosed efforts to recover back taxes from the period of noncompliance, and in some cases included government commitments to legalize short-term rentals or refrain from soliciting host information through the platform.¹³⁴ In effect, Airbnb traded a modest operational burden—automating tax collection at the point of booking—for substantial insulation from the zoning enforcement, registration requirements, and platform liability theories that posed existential

¹³¹ Andrew J. Bibler, Keith F. Teltser, Mark J. Tremblay, *Inferring Tax Compliance from Pass-Through: Evidence from Airbnb Tax Enforcement Agreements*, 103 REV. ECON. & STAT. 636 (2021).

¹³² *Airbnb, Inc. v. Schneiderman*, 989 N.Y.S.2d 786 (Sup. Ct. 2014) (upholding a modified subpoena compelling Airbnb to produce host data to the New York Attorney General for investigation of tax and zoning violations).

¹³³ *See Airbnb Spends More Than \$8 Million in San Francisco Fight*, *Time* (Nov. 3, 2015), <https://time.com/4096042/airbnb-san-francisco-proposition-f-campaign-rentals/> (reporting that Airbnb contributed \$8,377,784—over 90% of the opposition’s total war chest—to defeat Proposition F, which would have capped short-term rentals at 75 nights per year).

¹³⁴ *See, e.g., Elliot Njus, Airbnb, Acting as Portland’s Lodging Tax Collector, Won’t Hand Over Users’ Names or Addresses*, *Oregonian* (July 21, 2014), https://www.oregonlive.com/front-porch/2014/07/airbnb_acting_as_portlands_lod.html (describing Portland’s VCA as explicitly prohibiting the city from requiring Airbnb to disclose individually identifying host information).

risks to its business model.¹³⁵ The asymmetry of the exchange reflected the asymmetry of the parties' bargaining positions: cities needed revenue immediately and lacked the legal infrastructure to compel collection, while Airbnb could afford to wait out any individual city's enforcement efforts.

Airbnb did not merely sign VCAs to neutralize regulatory threats—it actively leveraged them as the centerpiece of a sophisticated public relations strategy. In December 2016, the company published a thirty-one-page "Policy Tool Chest" directed at municipal officials, presenting VCAs as a win-win that would "ensure that proper taxes are collected and remitted while relieving hosts of onerous tax filings and governments of the burden of collection and enforcement."^{136,137} The company promoted its tax collection figures aggressively, announcing that it had remitted over \$13.5 billion in tourism taxes to governments worldwide and framing itself as a partner in municipal governance rather than a target of it.¹³⁸ This messaging served a dual purpose. Externally, it positioned Airbnb as a responsible corporate actor in a "sharing economy" that empowered ordinary homeowners—a counternarrative to the mounting criticism that unregulated short-term rentals were displacing long-term residents and degrading neighborhoods. Internally, it provided political cover for city officials who signed VCAs, allowing them to point to new tax revenue as evidence that they had brought the platform to heel.

¹³⁵ See *Gannon v. Airbnb, Inc.*, 295 So. 3d 779 (Fla. Dist. Ct. App. 2020) (holding that Airbnb was not a "dealer" subject to direct hotel tax collection obligations under Florida law); see also Phil Williams, *REVEALED: Airbnb Legislation Shows How Capitol Hill Culture Affects Tennessee Communities*, *NewsChannel 5 Nashville* (Jan. 10, 2024), <https://www.newschannel5.com/news/newschannel-5-investigates/revealed-airbnb-legislation-shows-how-capitol-hill-culture-affects-tennessee-communities> (documenting Airbnb's lobbying campaigns to persuade state legislatures to preempt municipal short-term rental regulations).

¹³⁶ Airbnb, Inc., *Airbnb Policy Tool Chest* 5 (Dec. 2016), [https://www.airnbccitizen.com/wp-content/uploads/2016/12/National_PublicPolicyTool-ChestReport-v3.pdf](https://www.airbnb.com/content/uploads/2016/12/National_PublicPolicyTool-ChestReport-v3.pdf).

¹³⁷ *Id.* at 12 ("ensure that proper taxes are collected and remitted while relieving hosts of onerous tax filings and governments of the burden of collection and enforcement").

¹³⁸ Press Release, Airbnb, *Airbnb Has Generated More Than \$13.5 Billion in Tourism Taxes for Communities Worldwide* (Sept. 2023), <https://news.airbnb.com/airbnb-has-generated-more-than-13-5-billion-in-tourism-taxes/>.

Though there are differences between individual VCAs, they share some features in common. Each VCA is a bilateral contract, agreement, or written understanding signed by Airbnb and a government that stipulates conditions under which the firm will pay or assist in the collection of hotel taxes from a third party (a host) despite having no *legal obligation* to do so. Several of these agreements have been made public, and most share four types of clauses:¹³⁹

1. Tax liability. Airbnb will be held legally responsible for failure to report, collect, or remit hotel taxes.
2. Aggregate and/or anonymized auditing. The tax authority agrees to audit Airbnb based on its tax returns, rather than the individual hosts unless, and until an audit of Airbnb has been exhausted with the matter unresolved. Any host data reviewed by the tax authority will be anonymized.
3. Back taxes. The tax authority agrees not to pursue any action to recover unpaid taxes that had been due before the date the VCA went into effect.
4. Statute of limitations. Limits the amount of time in which Airbnb can be audited.

These agreements also frequently include concessions by the government entity, including but not limited to legalization of short-term rentals outside of designated areas and avowals not to solicit host-identifying information through Airbnb.¹⁴⁰

Airbnb implemented these collection agreements as follows. When a guest searches for a rental on Airbnb, she is presented with a set of search results that includes an image, location, and tax-exclusive estimate of the nightly fee for each.¹⁴¹ After a guest clicks

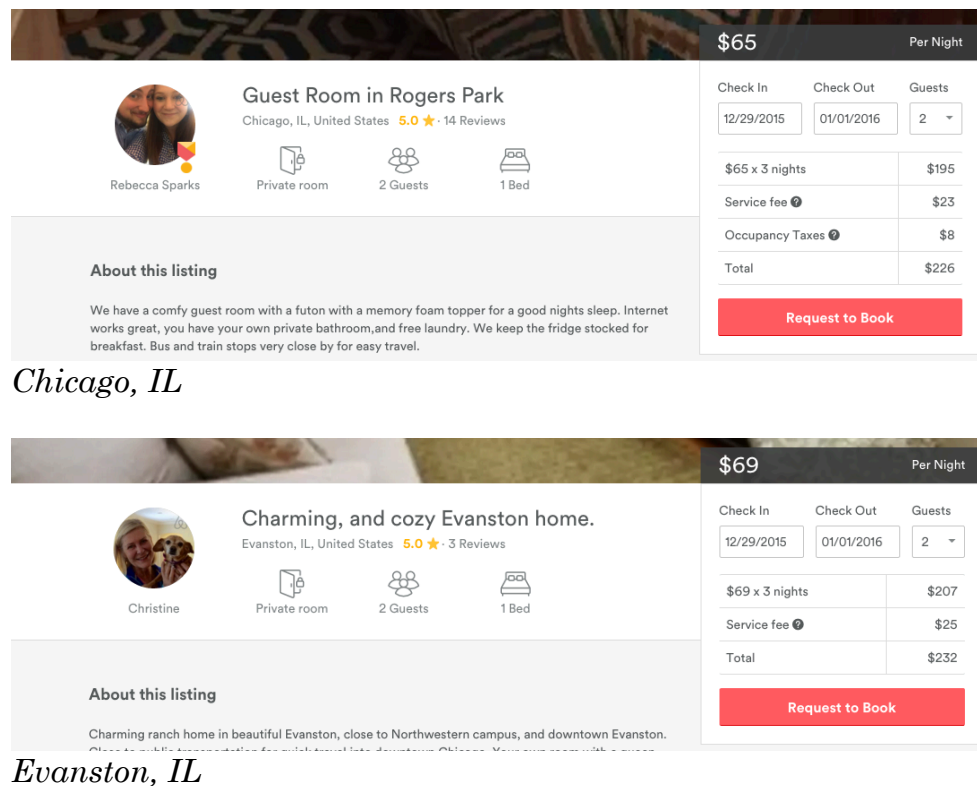
¹³⁹ See, e.g., VCAs for Sonoma County, CA, Mariposa County, CA, and Union City, CA.

¹⁴⁰ *Id.*

¹⁴¹ The price shown in the search results is the average cost per night of the room, excluding taxes and Airbnb's service fee. For example, if a listing's rental prices for Friday, Saturday, and Sunday are \$90, \$100, and \$110, respectively, and the listing has a \$30 cleaning fee, then the price displayed in the search results will be \$110 ($90+100+110+30 / 3 = 110$).

on a listing, she is shown a more detailed accounting of the rental cost, including Airbnb's service fee and any hotel tax. Figure 2 shows examples of listings from two jurisdictions: one that has a VCA with Airbnb (Chicago, IL), and one that does not (Evanston, IL). While not explicitly relevant for this Article, it is worth noting that Airbnb made no effort to make taxes salient to consumers—without clicking on a listing, it is not evident whether a hotel tax applies to it.

Figure 2. Airbnb listing details in Chicago and Evanston



Notes. This figure consists of two contemporaneous screenshots of listings in Chicago, IL (top) after it implemented a VCA and Evanston, IL (bottom) which had not done so. Notice that in Chicago, IL, Occupancy Taxes are listed as the third line item before the total amount due.

D. Did VCAs Work?

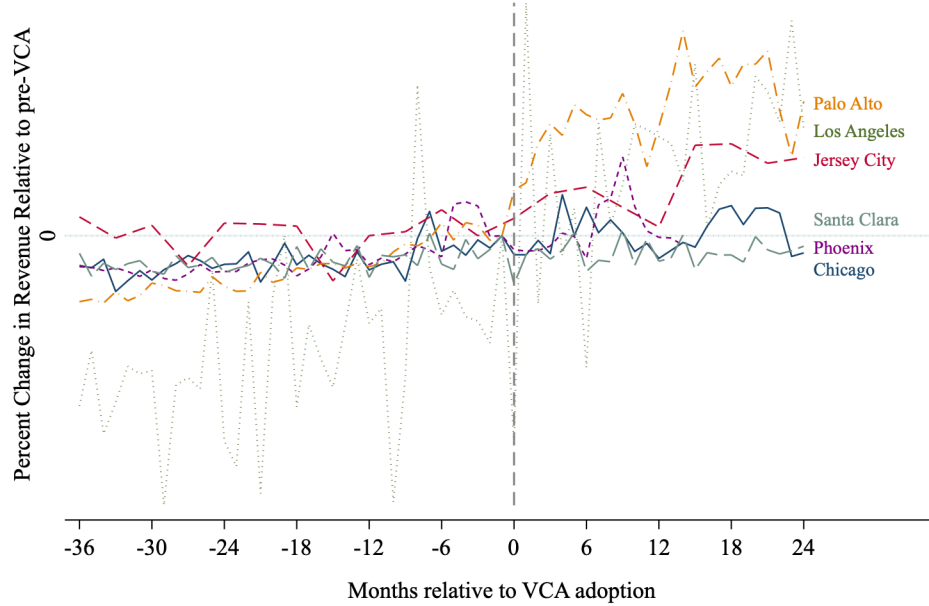
Armed with the context of VCA adoptions and their characterization as an imperfect but arguably necessary last resort by frustrated and ossified city regulators, the next logical question is to ask if they achieved their intended objective—did they actually increase city hotel tax revenues?

In this section, I evaluate the effects of the VCA on a city's total hotel tax receipts, using monthly tax collection data obtained from municipal governments primarily via Freedom of Information Act requests. By re-assigning the duty to remit hotel taxes from hosts to Airbnb, and therefore making it more difficult for hosts to evade the tax, the policy could be expected to have at least two effects on a city's hotel market and its hotel tax receipts. First, it effectively increases the price of Airbnb listings, and may therefore increase demand for hotel rooms to the degree that those are seen as substitutes for short-term rentals. Second, even if demand for Airbnb rentals declines somewhat following the policy, it will likely increase a city's hotel tax receipts as the opportunities for evasion dwindle.

1. A Natural Experiment

The core question of this research is whether cities that signed Voluntary Collection Agreements with Airbnb saw their hotel tax revenues change as a result. In some cases, the naïve comparison of raw revenues before the policy and after suggest a treatment effect. Figure 4 shows Palo Alto's revenues before and after the policy, demonstrating a clear trend break. Another descriptive rendering is shown in Figure 3 which plots log revenues for treatment cities normalized to event time equal to zero, i.e. VCA adoption. Note the significant heterogeneity by city.

But the empirical challenge is that hotel tax revenues are influenced by a variety of factors — local economies grow, tourism fluctuates with the business cycle, seasonal patterns shift — so it can be specious to simply compare revenues before and after a VCA and attribute the difference to the agreement itself.

Figure 3. Hotel Tax Revenues around VCA Adoption

Notes. This figure calculates revenues as a percentage of the revenue received in the period immediately prior to VCA adoption, where that amount has been normalized to zero. Time has also been normalized such that the month in which the VCA is adopted is plotted at zero. Includes seasonality fixed effects. This permits a direct comparison of revenues between cities and establishes the absence of pre trends in revenues—a vital assumption for the difference and difference estimates to follow.

Difference-in-differences estimation addresses this by using cities that never signed a VCA as a counterfactual. The logic works in two steps. First, you take the "difference" in revenue within each treated city before and after it adopted a VCA. But that difference captures everything that changed over time, not just the VCA. So the second "difference" subtracts out the change that occurred over the same period in control cities that didn't adopt a VCA. Whatever is left — the difference of the two differences — is the estimated causal effect of the agreement.

More formally, I estimate the following difference-in-differences specification:

$$y_{mt} = \gamma_m + \gamma_t + \pi Treat_m POST_{mt} + \varepsilon_{mt} \quad (1)$$

where y_{mt} is the outcome of interest for municipality m in month t . Characteristics invariant to municipality or time period are captured by municipality and time fixed effects, respectively. The coefficient of interest, π , captures the difference in the outcome between municipalities that adopted the policy and those that did not, both before and after its enactment.

For this estimation model to produce unbiased estimates, one condition must be met: parallel trends. The parallel trends assumption requires that in the absence of the policy, the prices in treated cities would have evolved similarly to control cities absent the policy. While this is a counterfactual that cannot be tested directly, we can observe whether the revenue trends evolve in parallel in the time periods leading up to the VCA, which provides suggestive evidence on how they would have evolved absent the policy change.

2. Dataset Construction and Descriptive Statistics

Individual taxpayer data that identifies the payer as an Airbnb owner is not disclosed by city tax authorities due to confidentiality restrictions. To measure the effects of the VCAs on cities' tax revenue, I constructed a panel dataset of aggregate city-level hotel tax collections for thirteen cities—ten that implemented VCAs in years 2014-2016 ("treated cities") and three that did not ("control cities"). For two cities—Santa Clara and Los Angeles—historical information on monthly hotel revenue is available through publicly available budget documents and reports. Data for the eleven other cities was obtained via Freedom of Information Act requests.¹⁴² Most of the cities for which data was obtained track revenue monthly, but Malibu and Jersey City track revenues on a quarterly

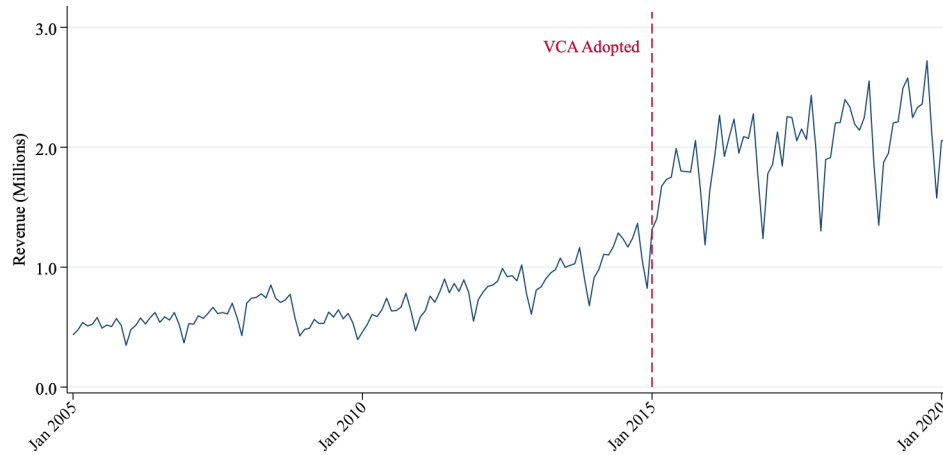
¹⁴² The federal Freedom of Information Act, 5 U.S.C. § 552 (2018), applies only to federal agencies. Each state has enacted its own public records law—variously styled as "freedom of information," "open records," "sunshine," or "public records" acts—that governs requests to state and local government entities. See Reporters Comm. for Freedom of the Press, *Open Government Guide*, <https://www.rcfp.org/open-government-guide/> (last visited Apr. 30, 2026) (compiling and comparing the public records laws of all fifty states and the District of Columbia). The data used in this study were obtained under the applicable state or local public records statute in each jurisdiction.

basis. The time periods over which data was available differed widely by city, with some cities providing up to a decade's worth of data and others limited to only a few years surrounding the date of enactment. All cities, along with their VCA implementation months, are listed in Table 1.

Table 1. Hotel Tax Revenue Descriptive Statistics

City	VCA Date	N	Descriptive Statistics			
			Mean	SD	Median	Tax Rate (%)
Chicago, IL	Feb 2015	170	7.375	3.771	6.413	4.5
Dallas, TX	—	101	4.669	1.037	4.668	—
Jersey City, NJ	Feb 2016	61	1.495	0.697	1.454	6.0
Los Angeles, CA	Aug 2016	92	20.382	6.718	19.045	14.0
Malibu, CA	Apr 2015	59	0.323	0.170	0.292	12.0
Palo Alto, CA	Jan 2015	248	0.999	0.627	0.728	14.0
Philadelphia, PA	Jul 2015	61	5.630	1.391	5.772	8.5
Phoenix, AZ	Jul 2015	135	2.764	1.052	2.529	5.0
Portland, OR	Jul 2014	32	7.929	7.883	3.171	—
Santa Clara, CA	Nov 2015	170	1.244	0.877	0.999	9.5
Savannah, GA	—	62	1.795	0.382	1.772	—
Washington, DC	Feb 2015	105	17.428	4.155	17.410	14.5
All Cities		1,296	5.464	6.782	2.500	

Notes. This table includes key policy details and descriptive statistics on each cities hotel tax revenues (in millions). Data from the following cities was obtained via Freedom of Information Act requests: Chicago, Dallas, Jersey City, Malibu, Palo Alto, Philadelphia, Portland, Savannah and Washington D.C. The cities of Los Angeles and Santa Clara make this data publicly available.

Figure 4. Raw Revenue Data for Palo Alto, CA

Notes. This figure plots the raw revenue reported by the city of Palo Alto in response to a FOIA request for years 2005-2020. Palo Alto was chosen because it has modest total revenues but a comparatively high market share for Airbnb relative to hotels and data covering a long pre-period. The red vertical line is drawn on the date that Palo Alto’s VCA with Airbnb went into effect. There is clear seasonality in revenue receipt, but the vertical line appears to represent a sudden shift in the *level* of revenues.

3. Regression Results

Table 2 reports results for four specifications based on Equation 1, which differ in functional form, fixed effects and estimator. The first two columns regress revenue in levels (millions of dollars) on the binary VCA indicator. Column 1 reports the most parsimonious two way fixed effect (TWFE) specification, including only city fixed effects—it asks whether treated cities collected more revenue after the VCA than before, relative to control cities, without absorbing common time shocks. Column 2 adds month-year fixed effects to the levels specification, which absorbs common macroeconomic and seasonal shocks e.g. recessions, tourism trends, etc.) that hit all cities simultaneously. While both estimates of the treatment effect are positive, they are not statistically significant at conventional thresholds. However, this lack of significance largely reflects the enormous variation in revenue levels across cities; a million-dollar

swing in Palo Alto means something very different than in Los Angeles, and pooling observations in levels inflates the residual variance.

To address this limitation, Columns 3 and 4 switch the dependent variable to log revenue, keeping both city and month-year fixed effects. In using logs, these specifications effectively estimate percentage changes, which normalizes different sized cities to a comparable scale. Column 3 reports the preferred estimate, reflecting both appropriate scaling and the netting out of seasonality and common time trends. The coefficient estimate is 0.141 (SE = 0.049, $p = 0.023$) — statistically significant at the 5% level, which translates into a roughly 14% increase in hotel tax revenue.

Finally, Column 4 reports results from using the Callaway & Sant'Anna (2021) estimator on log revenue. This is an important robustness check given concerns about two-way fixed effect estimators in a setting with staggered treatment timing: if treatment effects are heterogeneous across cities or evolve over time—both plausible in this context, the TWFE estimate can be biased because already-treated cities serve as implicit controls for later-treated ones. The Callaway & Sant'Anna estimator avoids this by constructing proper group-time average treatment effects using only never-treated or not-yet-treated units as controls. This model drops the quarterly cities (Jersey City and Malibu) to maintain strict panel balance. The average treatment effect on the treated (ATT) estimate is 0.127 (SE = 0.053, $p = 0.017$) — also significant, and reassuringly close to the TWFE estimate, suggesting that bias from staggered adoption is not a major concern.

When considered together, the difference in differences estimates indicate a positive, detectable effect of the VCA on revenues when the outcome variable is scaled correctly to address cross city heterogeneity in revenue magnitudes (i.e. in logs rather than levels.) While the level specifications (Col. 1 and 2) lack power, the log specifications (Col. 3 and 4) both find a statistically significant positive effect of roughly 12–15%, and the robustness of this finding to the CS correction strengthens the causal interpretation.

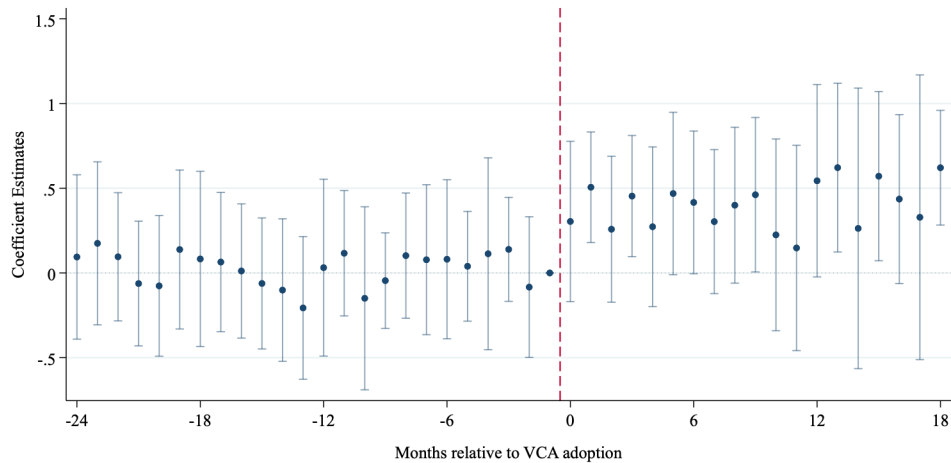
In addition to the difference in difference estimates, I also report estimates for the event-study analogue of the Col. 3

specification in Figure 4. In contrast to the TWFE estimates, which provide an average treatment effect across all time periods, the event study coefficients are dynamic, illustrating how log hotel tax revenue in treated cities evolved relative to control cities in the months surrounding VCA adoption. The pre-treatment coefficients ($t = -24$ through $t = -2$) are clustered around zero, lending credibility to the parallel trends assumption—i.e., that the treated and control cities were on similar revenue trajectories before the VCA and thus the control cities are an appropriate counterfactual. The post-treatment coefficients ($t = 0$ onward) estimate the period-by-period effect of VCA adoption on log revenue. The marked increase in coefficient values after the policy supports the conclusion that the VCA immediately increased hotel tax collections, bringing in revenue that was previously uncollected.

Table 2. Effect of VCA on Hotel Tax Revenues

	Estimated Effect on			
	(1)	(2)	(3)	(4)
	Revenue	Revenue	ln(Revenue)	ln(Revenue)
VCA	1.448 (0.883)	1.062 (0.980)	0.141** (0.049)	0.127** (0.053)
Observations	1,236	1,236	1,236	986
Adj. R-squared	0.857	0.845	0.949	—
City FE	Yes	Yes	Yes	—
Month-Year FE	No	Yes	Yes	—
Estimator	TWFE	TWFE	TWFE	CS (2021)

Notes. Standard errors clustered at the city level in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (4) uses the Callaway & Sant'Anna (2021) estimator with never-treated as control. CS event-study pre-treatment avg: -0.0096^{**} (SE = 0.0029); post-treatment avg: $.2097^{**}$. (SE = .0794.). Quarterly cities (Jersey City, Malibu) excluded from column (4), the CS estimator, because of strong balanced panel requirements.

Figure 5. Event Study Estimates of Log Revenues on VCAs

Notes. This figure plots the coefficient estimates of the effect of VCA adoption on log hotel tax revenue. These estimates are generated from an event study specification analogous to DID Model 3 from Table X *supra* and include city and month fixed effects to control for seasonality.

D. Buyer's Remorse

Despite anecdotal and quantitative evidence (discussed in Section II.C) that Airbnb VCAs were effective in their primary objective of collecting taxes, some cities with VCAs are now seeking to nullify or withdraw from them in favor of more restrictive legislation. In this section, I argue that many, though not all, of the ongoing problems and disillusionment with VCAs stem from provisions limiting cities' access to individual host information.

Difficulty Auditing Transactions. Despite their wide adoption, and preliminary evidence that they are effective at increasing revenue collections, some cities are reconsidering their VCAs. For example, San Diego City Council is attempting to pass ordinances that

would override its VCA,¹⁴³ citing stonewalling by Airbnb in response to questions about irregularities in remittances.

Quality of Life Regulations. Although it is problematic to verify whether Airbnb is remitting all taxes to the city without individual transaction data, by far the most significant consequence of cities' forfeiture of host identifying and transaction data is the inability to enforce various *non-tax* regulations.

For example, New Orleans, despite being viewed as the “poster child” of the VCA effort in 2016,¹⁴⁴ claims that Airbnb reneged on a key part of their agreement, intended to set up a host registration system. After years of back and forth, the city claims that Airbnb failed to hand over any of the necessary data (including, for example, address or account owner name or contact information). New Orleans apparently abandoned the effort, passing one of the most restrictive regulations in the country that limit licenses to rent to one property per block and a requirement that the licensee live on the property.¹⁴⁵ Whether they can enforce these regulations without a registry is uncertain.

Lack of a functional registry also prevents local tax authorities from recouping taxes owed on previous transactions from hosts directly, and prevents them from monitoring their behavior on other platforms, including direct competitors to Airbnb, that have not signed VCAs. Put differently, VCAs can make local governments permanently dependent on the individual firm for significant revenues, and are signed when those firms have accrued sufficient market power to negotiate them on their terms.

Historically, hosts on Airbnb have generally been unaware of their hotel tax liability and, as a byproduct, of their obligation to

¹⁴³ See Voluntary Collection Agreement for Sonoma County Transient Occupancy Tax, 1–2 (2016); Paris Martineau, Inside Airbnb's “Guerrilla War” Against Local Governments, *Wired*, <https://www.wired.com/story/inside-airbnbs-guerrilla-war-against-local-governments/> (Mar. 20, 2019).

¹⁴⁴ Katie Benner, *New Orleans Becomes New Model for Airbnb to Work With Cities*, *N.Y. Times* (Oct. 7, 2016), <https://www.nytimes.com/2016/12/07/technology/new-orleans-airbnb-model.html>.

¹⁴⁵ Kevin McGill, *New Orleans Attempts Again to Regulate Vacation Rentals*, *AP News* (Mar. 24, 2023, 6:11 PM), <https://apnews.com/article/new-orleans-short-term-rental-regulations-a53e5f391ed86cd475d4fe3b9f6f06e9>.

remit tax payments to the local government.¹⁴⁶ This lack of awareness of, and preparedness to comply with, hotel taxes was mutual on the part of cities. Most cities' systems for accepting hotel tax payments were not built for or prepared to accept payments from a flood of amateur STR operators. The forms that taxpayers are required to complete are overly complicated, and the methods used to make payment are generally not user-friendly.

There are two additional areas in which the inaccessibility of identifying data made regulation difficult.

An end run around residential zoning. Most cities have separate zones for residential and commercial activity, with the latter sometimes including zones specifically for hotels and entertainment. In some cities, STRs are permitted in owner-occupied housing if they do not exceed a certain number of days per year (usually 14).¹⁴⁷ If an Airbnb host does not comply with these regulations, then they effectively perform an end run around residential zoning. In doing so, the host derives an advantage that is unavailable to traditional hotels—the ability to offer STRs in residential locations where consumers may prefer to stay. Yet such non-compliant residential STRs likely create several externalities as well, including higher housing and rental costs, noise complaints, and security issues.

Consumer Safety Regulations. Hotels are regulated to protect the guests who stay in them. Because hotel guests are often not local, they may face asymmetric information that would make it difficult to ascertain the quality of a hotel before choosing to stay in it. Hotel regulations are meant to address this potential market failure by establishing a floor for hotel quality: cleanliness, security, building codes (including fire alarms), accessibility for the disabled, and insurance.¹⁴⁸

¹⁴⁶ Shu-Yi Oei & Diane M. Ring, *Can Sharing Be Taxed?*, 93 WASH. U.L. REV. 989, 1016 (2016).

¹⁴⁷ Lodgify, *Airbnb & Short-Term Rental Rules and Regulations in California* (2024).

¹⁴⁸ Omland, *Key Steps To Stay Compliant With Health And Safety Regulations In The Hospitality Sector* (Aug. 25, 2023), <https://www.omlandhospitality.com/blog/hotel-safety-regulations-management-guide>.

In contrast, as with hotel taxes, hosts on Airbnb have generally been unaware of their need to comply with consumer safety regulations that pertain to STRs. Airbnb has also taken steps to stymie efforts by local governments to bring hosts' STRs into compliance. For example, in 2014, Portland, OR asked Airbnb to register each host with the city so that the city could ensure that each STR had a fire extinguisher in or near the kitchen. Airbnb declined to provide hosts' addresses to the city, fearing that inspectors would be perceived as invasive and would hamper Airbnb's growth in the city.

III. POLICY IMPLICATIONS

In the previous section, I argued that VCAs that leave cities without access to individual level transaction data are a flawed tool for tax collection. However, there are circumstances in which a VCA, despite its limitations, is still the best weapon in the government's arsenal. In this section, I discuss factors that cities should consider in determining whether to pursue a VCA with a platform firm and what proactive steps policy makers might take to improve leverage to secure more favorable terms than those common to Airbnb VCAs. Finally, I situate the lessons from Airbnb VCAs and the role of VCAs generally in the broader policy debate about the limits of information reporting without remittance.

A. *An Imperfect Solution*

Cities must weigh the short-term revenue benefits of a VCA against the long-term costs documented in Section II.D., *supra*. A VCA is, even at maximum efficacy, a second-best instrument—it sacrifices transparency and regulatory capacity in exchange for immediate collections. But second-best instruments exist for a reason. When the alternative is no collection at all, a VCA may still represent the best available policy option.¹⁴⁹ Whether it does depends on several factors specific to the city's circumstances.

¹⁴⁹ Mary Riddel & Pavel Yakovlev, *Inferring Tax Compliance from Pass-Through: Evidence from Airbnb Tax Enforcement Agreements*, 103 REV. ECON. & STAT. 636, 636 (2021).

The first is the nature of the city's short-term rental market. A VCA is most defensible when a city's primary concern is capturing revenue rather than managing residential displacement. In tourism-dependent jurisdictions where short-term rentals are concentrated in resort or vacation properties—rather than in residential neighborhoods—the data-access concessions embedded in VCAs are less costly.¹⁵⁰ Florida's experience is instructive: between 2015 and 2017, Airbnb signed bed-tax agreements with thirty-nine of the state's sixty-three counties that assess such a tax, and the company's statewide remittances roughly doubled from \$20 million to \$45.7 million over that period.¹⁵¹ For smaller resort communities like Charlotte County or Pinellas County, where short-term rentals largely substitute for hotel stays rather than displacing long-term tenants, this new revenue arrived with few of the residential welfare concerns that plagued cities like New Orleans and San Francisco. By contrast, cities where the primary policy challenge is the conversion of residential housing stock to de facto hotels—a concern disproportionately concentrated in high-cost urban markets—trade away the very regulatory tools they need most when they sign a VCA without data-sharing provisions.¹⁵²

The second factor is institutional capacity. A VCA is more justifiable when a city's tax administration is too under-resourced to enforce collection through conventional means. Prior to VCAs, voluntary compliance rates for short-term rental taxes were strikingly low—Riddel and Yakovlev estimate that taxes were paid on at most twenty-four percent of Airbnb transactions before

¹⁵⁰ In tourism-dependent resort communities, short-term rentals largely substitute for hotel stays rather than displace long-term tenants, making revenue collection—rather than residential enforcement—the primary policy objective.

¹⁵¹ David Ashton, *Airbnb Remitted \$45.7 Million in Taxes in Florida in 2017*, *Fla. Trend* (Feb. 8, 2018), <https://www.floridatrend.com/article/23914/airbnb-remitted-457-million-in-taxes-in-florida-in-2017>.

¹⁵² See Paris Martineau, *Inside Airbnb's Guerrilla War Against Local Governments*, *Wired* (Mar. 20, 2019), <https://www.wired.com/story/inside-airbnbs-guerrilla-war-against-local-governments/> (documenting how cities that traded data access for tax revenue found themselves unable to enforce zoning and registration requirements).

enforcement agreements took effect.¹⁵³ This compliance gap reflects the near-impossibility of municipal tax offices, many of which still reconcile lodging tax collections manually, identifying and auditing thousands of individual hosts operating without permits or registration.¹⁵⁴ For cities that lack the staff, technology, and legal infrastructure to build independent enforcement regimes, a VCA represents a pragmatic concession: imperfect collection through the platform is preferable to negligible collection without it. The risk, however, is that the arrangement becomes self-reinforcing. By outsourcing collection to the platform without developing independent capacity, cities remain dependent on the platform's cooperation and unable to verify the accuracy of aggregate remittances.¹⁵⁵

Third, the political environment matters—and in particular, whether the city faces hostile state preemption of local regulatory authority. Several state legislatures have acted to curtail or eliminate municipal power to regulate short-term rentals, often at the urging of the platform industry. Arizona's SB 1350, enacted in 2016, prohibited cities and counties from banning short-term rentals based on rental duration, voiding local restrictions that cities like Sedona and Scottsdale had attempted to impose.¹⁵⁶¹⁵⁷ Florida's legislature banned new local vacation rental rules almost entirely in 2011, and although it partially reversed course in 2014, the state has continued to push toward statewide standardization

¹⁵³ Riddel & Yakovlev, *supra* note 1, at 636 (estimating that taxes were paid on at most twenty-four percent of Airbnb transactions before enforcement agreements took effect).

¹⁵⁴ See Darien Shanske, *Public Finance Considerations for Short-Term Rental Tax Regulation*, 65 STATE TAX NOTES 709, 712 (2012) (describing the administrative burden on local tax offices attempting to identify and audit individual short-term rental hosts).

¹⁵⁵ See Richard Pomp, *Airbnb and Its Voluntary Tax Collection Efforts: Myth vs. Reality* 5–7 (Univ. of Conn. Sch. of L., Working Paper, 2019) (arguing that aggregate remittances without property-level data make independent verification impossible and create a structural dependency on the platform's cooperation).

¹⁵⁶ S.B. 1350, 52d Leg., 2d Reg. Sess. (Ariz. 2016) (codified at Ariz. Rev. Stat. Ann. § 9-500.39 (2016)).

¹⁵⁷ See Michael Marek, *Arizona's Short-Term Rental Dilemma*, 55 ARIZ. ST. L.J. 891 (2024) (analyzing the continuing conflict between municipalities and the state legislature over local regulatory authority).

at the expense of local control.¹⁵⁸ In this environment, a VCA may be the only instrument a city can deploy without risking further legislative backlash. Where state government is sympathetic to the platform's position—or actively aligned with it—a city that attempts to impose registration requirements, data-sharing mandates, or comprehensive licensing regimes may provoke preemptive legislation that strips away regulatory tools entirely. A VCA, by contrast, is a voluntary agreement that a platform has affirmatively chosen to sign, making it more politically durable even in hostile state-level environments.¹⁵⁹

Fourth, the structure of the platform market affects the calculus. A VCA is most effective when a single platform dominates the local short-term rental market, because a bilateral agreement with one firm can capture the vast majority of transactions. In the mid-2010s, Airbnb held a commanding share of urban short-term rental bookings in the United States, particularly in the major metropolitan markets where VCAs were first adopted.¹⁶⁰ But in vacation and resort markets, VRBO—then HomeAway, acquired by Expedia in late 2015 for \$3.9 billion—maintained a substantial presence.¹⁶¹ Because VCAs are bilateral, a city that signs with Airbnb but not VRBO collects taxes on only a fraction of short-term rental activity while giving hosts an incentive to shift bookings to the uncovered platform. Several jurisdictions experienced exactly this gap: in Palm Desert, California, for example, Airbnb collected the transient occupancy tax while VRBO hosts remained

¹⁵⁸ Fla. Stat. Ann. § 509.032(7) (West 2011), *amended by* Act of June 20, 2014, ch. 2014-71, 2014 Fla. Laws 1 (narrowing preemption to duration and frequency restrictions). *See also* S.B. 1537, 2024 Leg., Reg. Sess. (Fla. 2024) (further shifting toward statewide standardization at the expense of local control).

¹⁵⁹ *Cf.* Richard C. Schragger, *The Attack on American Cities*, 96 TEX. L. REV. 1163, 1200–05 (2018) (analyzing preemption as a tool used by state legislatures to override municipal regulatory preferences, particularly where industry lobbying is concentrated at the state level).

¹⁶⁰ By September 2016, Airbnb had approximately 2.8 million global listings compared to HomeAway's roughly 1.2 million. *See* Brad Stone, *THE UPSTARTS: HOW UBER, AIRBNB, AND THE KILLER COMPANIES OF THE NEW SILICON VALLEY ARE CHANGING THE WORLD* 290–95 (2017).

¹⁶¹ Expedia, Inc. acquired HomeAway, Inc. (parent of VRBO) in November 2015 for \$3.9 billion, closing in the first quarter of 2016. *See* Greg Bensinger, *Expedia to Buy HomeAway for \$3.9 Billion*, *Wall St. J.* (Nov. 4, 2015), <https://www.wsj.com/articles/expedia-to-buy-homeaway-for-3-9-billion-1446636068>.

responsible for self-remittance.¹⁶² The more fragmented the platform market, the less a single VCA can accomplish—and the stronger the case for a regulatory framework that applies uniformly to all platforms regardless of whether they have volunteered to participate.

These four conditions suggest a rough taxonomy. Cities that are small, tourism-dependent, institutionally constrained, operating under hostile state preemption, and served primarily by a single dominant platform have the strongest case for treating a VCA as the best available option. Cities at the opposite end of each dimension—large, residentially concerned, institutionally capable, politically autonomous, and facing a fragmented platform market—have the weakest case, and should instead pursue comprehensive regulatory frameworks that preserve data access and apply uniformly across platforms.¹⁶³¹⁶⁴ The difficulty, of course, is that most cities fall somewhere in between, and the conditions are not static. A city that signs a VCA under favorable conditions may find, as New Orleans and Chicago discovered, that its short-term rental market evolves in ways that make the agreement's concessions increasingly costly over time.¹⁶⁵¹⁶⁶

¹⁶² See Avalara MyLodgeTax, *Short-Term Rental Tax Collection: Platform Gaps and Compliance Challenges* (2020) (documenting jurisdictions where Airbnb collects transient occupancy taxes but VRBO hosts remain responsible for self-remittance).

¹⁶³ Denver, Colo., Ordinance No. 473 (June 6, 2016) (requiring that short-term rentals be the host's primary residence and establishing a mandatory licensing regime).

¹⁶⁴ Boston, Mass., Ordinance Regulating Short-Term Rentals (June 13, 2018) (requiring owner-occupancy, registration, and platform data-sharing provisions). See also Sarah Yaussi, *Regulating Short-Term Rentals: A City Guide*, NAT'L LEAGUE OF CITIES 18–20 (2019) (identifying Boston as a model for preserving housing stock while capturing revenue).

¹⁶⁵ See Richard A. Webster, *New Orleans Has New Short-Term Rental Rules. Here's What You Need to Know*, *Times-Picayune* (Dec. 1, 2019) (reporting that approximately 5,710 listings were active compared to only 1,260 permits in circulation). See also Kyle Barron, Edward Kung & Davide Proserpio, *The Effect of Home-Sharing on House Prices and Rents: Evidence from Airbnb*, 40 MKTG. SCI. 23 (2021) (analyzing the effects of Airbnb's growth on housing markets).

¹⁶⁶ See Chanwool Kim & Liyuan Wang, *The Effects of Short-Term Rental Regulation: Insights from Chicago*, J. HOUS. ECON., Mar. 2024 (finding that Airbnb listings fell approximately sixteen percent following the introduction of regulatory requirements).

B. Increasing Governments' Leverage

The analysis in Part II and Section III.A suggests that many cities signed VCAs from a position of weakness—confronting a well-resourced platform with dedicated lobbying teams while operating with limited legal budgets and fragmented regulatory authority.¹⁶⁷¹⁶⁸ The bilateral structure of VCAs compounded this disadvantage. Each city negotiated individually, often subject to confidentiality clauses that prevented it from sharing the agreement's terms with other jurisdictions.¹⁶⁹ The result was a classic information asymmetry: Airbnb knew the terms of every VCA it had signed, while no city knew what any other city had agreed to. As cities look to the future and the likelihood of further platform disruption in other industries, several strategies could help rebalance this dynamic.

The most direct response to bilateral asymmetry is collective bargaining. If several large cities agreed to negotiate with a platform jointly, they could pool legal resources, share information about the platform's negotiating positions, and present a unified set of minimum terms. The European experience offers a proof of concept. In 2019, a coalition of twenty-two European cities—including Amsterdam, Barcelona, Berlin, Paris, and Vienna—

¹⁶⁷ See Richard Pomp, *Airbnb and Its Voluntary Tax Collection Efforts: Myth vs. Reality* 3–5 (Univ. of Conn. Sch. of L., Working Paper, 2019), https://digitalcommons.lib.uconn.edu/law_papers/560/ (documenting the bilateral structure and confidentiality provisions of Airbnb's voluntary collection agreements).

¹⁶⁸ Sharing Cities Action, *Short-Term Rentals: Data, Negotiation and Collaboration Strategies for Cities* 12 (2020), <https://www.sharingcitiesaction.net/wp-content/uploads/2020/06/Short-Term-Rentals-Data-Negotiation-and-Collaboration-Strategies-for-Cities-Report.pdf> [hereinafter *Sharing Cities Report*] (concluding that "[n]one of the cities have reached a satisfactory end as a result of their negotiation efforts" and attributing this to platforms' "specialized team, both at the headquarters and at the country level, to deal with a multi-layered lobby strategy").

¹⁶⁹ See Voluntary Collection Agreement Between City of Santa Fe and Airbnb, Inc. § 8 (requiring Airbnb's "prior written authorization" before the city could release any information about the agreement "in the press, on the internet or any other form of media"). See also Pomp, *supra* note 103, at 6–8 (analyzing how confidentiality clauses prevent cities from comparing terms or building on negotiated precedents).

coordinated to demand that the European Union require platforms to share host-level data with local authorities.¹⁷⁰ Their joint advocacy contributed directly to the enactment of EU Regulation 2024/1028, which requires all short-term rental hosts to register with national authorities, obtain unique identification numbers, and mandates that platforms transmit activity data—nights rented, guest counts, and property addresses—monthly to a centralized national portal.¹⁷¹ No single European city could have extracted these concessions bilaterally. The coalition succeeded precisely because it shifted the forum from bilateral negotiation to legislative advocacy—leveraging collective political weight to secure data-sharing requirements that no individual VCA had achieved. A comparable coalition of major American cities—New York, Los Angeles, San Francisco, Chicago, and Boston, for example—could exert similar pressure, whether through joint negotiation, coordinated litigation, or advocacy for federal marketplace regulation.

Even short of formal collective bargaining, institutionalized information-sharing between cities could help level the playing field. Cooperative monitoring would allow cities to identify emerging platforms operating in regulatory gray areas before those platforms accumulate the political constituencies and market share that make regulation difficult. The National League of Cities has recommended that municipalities develop shared enforcement frameworks and peer learning networks to reduce duplication of effort.¹⁷² Interlocal cooperation agreements—already a well-

¹⁷⁰ In 2019, a coalition of twenty-two European cities—including Amsterdam, Barcelona, Berlin, Paris, and Vienna—issued a joint letter demanding that the European Union require platforms to share host-level data with local authorities. See Martina Schroeter & Maureen Mukamayanja, *The EU's Transformative Short-Term Rental Regulation*, REG. REV. (Mar. 11, 2026) (tracing the coalition's lobbying efforts and their influence on the EU's regulatory response).

¹⁷¹ Council Regulation 2024/1028 of Apr. 11, 2024, on the Collection and Sharing of Data Relating to Short-Term Accommodation Rental Services, 2024 O.J. (L 1028) 1 (requiring all short-term rental hosts to register with national or local authorities, obtain unique identification numbers, and mandating that platforms transmit activity data—including nights rented, guest counts, and property addresses—monthly to a national "Single Digital Entry Point").

¹⁷² See NLC Guide, *supra* note 100, at 18–20 (recommending that municipalities develop "shared enforcement frameworks" and "peer learning

established tool in areas like law enforcement, code enforcement, and joint purchasing—provide a ready-made legal framework. The Sharing Cities Action network, a global collaboration launched in 2019 in partnership with the Inside Airbnb data project, has begun to fill this role, drawing on experiences from twenty-four cities across fifteen countries to compile data on platform practices, VCA terms, and enforcement outcomes.¹⁷³ But its central finding is sobering: despite years of coordination, none of the participating cities reported a satisfactory negotiation outcome with Airbnb, underscoring the depth of the asymmetry that cooperative monitoring alone cannot fully overcome.¹⁷⁴

A third strategy is standardized data-sharing requirements that apply uniformly across jurisdictions. One of Airbnb's most effective arguments against mandatory collection and reporting was that hotel tax regulations vary across thousands of municipalities with different rate structures, definitions of short-term stays, and exemptions—making compliance prohibitively burdensome. Standardization undercuts this objection.¹⁷⁵ California took a significant step in this direction with SB 346, effective January 1, 2026, which authorizes local governments statewide to compel platforms to share host names, addresses, nights booked, and registration status—the first state law creating a statutory data-sharing mandate rather than relying on voluntary

networks" to reduce duplication of effort). *See also* Wash. Mun. Rsch. & Servs. Ctr., *Interlocal Cooperation Basics* (2023), <https://mrsc.org/explore-topics/government-organization/interlocal/interlocal-cooperation-basics> (describing the statutory framework for interlocal agreements for joint service delivery, shared personnel, and enforcement cooperation).

¹⁷³ Sharing Cities Report, *supra* note 104, at 25 ("Cities need to cooperate, exchange information and coordinate actions to counterbalance the asymmetric power of global platforms.").

¹⁷⁴ *See* Inside Airbnb, *Platform Failures* (2024), <https://insideairbnb.com/report/platform-failures/> (documenting that in Barcelona, sixty to seventy percent of host addresses in platform-provided data were missing or incorrect, and that in Berlin and Paris, eighty and sixty percent of listings, respectively, remained illegal).

¹⁷⁵ *Cf.* Darien Shanske, *A New Theory of the State Corporate Income Tax: The State Corporate Income Tax as Retail Sales Tax Complement*, 66 TAX L. REV. 305, 340–45 (2013) (arguing that uniformity in tax base definitions across jurisdictions reduces platform compliance costs and eliminates a key objection to mandatory collection obligations).

agreements.¹⁷⁶ Cities like Los Angeles and San Francisco had already implemented their own registration and data-reporting regimes, but the patchwork of local requirements imposed real compliance costs and gave platforms a credible argument against cooperation.¹⁷⁷ A uniform statewide standard eliminates this objection while ensuring that all jurisdictions within the state have access to the same enforcement tools, regardless of whether they have the resources to negotiate individually with each platform.

Finally, cities may benefit from delegating negotiation and enforcement to state governments that share their regulatory objectives. Several states have adopted marketplace facilitator laws that require platforms to collect and remit lodging taxes statewide, removing the need for city-by-city VCAs altogether. Oregon and Colorado have imposed marketplace facilitator obligations on short-term rental platforms.¹⁷⁸ Colorado has gone further, directing the state property tax administrator to establish a pilot statewide database and uniform reporting system for short-term rental units.¹⁷⁹ Hawaii's approach is perhaps the most comprehensive: its 2024 legislation expanded county zoning powers while requiring platforms to register with counties, pay registration fees, and file monthly activity reports—combining statewide tax collection with local land-use authority.¹⁸⁰ The risk

¹⁷⁶ S.B. 346, 2023–2024 Reg. Sess. (Cal. 2024) (effective Jan. 1, 2026) (authorizing local governments to compel short-term rental platforms to share host name, address, nights booked per year, and registration status).

¹⁷⁷ L.A., Cal., Ordinance No. 185,931 (Dec. 11, 2018) (requiring hosts to register, display registration numbers on all listings, and retain booking records for three years). *See also* S.F., Cal., Admin. Code ch. 41A (establishing the Office of Short-Term Rentals and requiring platforms to file quarterly reports showing all San Francisco listings, nights booked, and host registration status).

¹⁷⁸ H.B. 4134, 81st Leg. Assemb., Reg. Sess. (Or. 2022) (raising the statewide transient lodging tax from 1.5% to 2.75% and imposing marketplace facilitator obligations on platforms). *See also* Colo. H.B. 19-1240, 72d Gen. Assemb., 1st Reg. Sess. (2019) (assigning sales tax collection to "marketplace facilitators," including short-term rental platforms).

¹⁷⁹ S.B. 24-033, 74th Gen. Assemb., 2d Reg. Sess. (Colo. 2024) (directing the property tax administrator to establish a pilot statewide database and uniform reporting system for short-term rental units).

¹⁸⁰ S.B. 2919, 32d Leg., Reg. Sess. (Haw. 2024) (expanding county zoning powers over short-term rentals, requiring platforms to register with counties, pay a \$1,000 fee, and file monthly reports, while maintaining a statewide Transient Accommodations Tax of eleven percent).

of state-level delegation, however, is the one documented in Section II.A: where state governments are sympathetic to platform interests, delegation may result in preemption of local authority rather than augmentation of it.¹⁸¹ The viability of this strategy therefore depends on political alignment between city and state—a condition that, as discussed, is far from universal.

CONCLUSION

This Article has presented what is, to the author's knowledge, the first causal evidence on the revenue effects of Voluntary Collection Agreements between a platform firm and municipal governments. Using original panel data on monthly hotel tax collections obtained primarily through public records requests, and exploiting the staggered adoption of VCAs across cities in a difference-in-differences framework, I find that VCAs increased hotel tax revenues by approximately twelve to fifteen percent—an estimate that is robust to the Callaway and Sant'Anna estimator designed to address bias from heterogeneous treatment effects under staggered timing.¹⁸²¹⁸³ This is a meaningful result. Prior to VCAs, taxes were paid on at most a quarter of Airbnb transactions,¹⁸⁴ and the event-study estimates confirm that the revenue gains materialized immediately upon adoption, with pre-treatment coefficients clustered around zero lending credibility to the parallel trends assumption. Whatever else may be said about

¹⁸¹ See Richard C. Schragger, *The Attack on American Cities*, 96 TEX. L. REV. 1163, 1200–05 (2018) (documenting the risk that state-level delegation invites industry lobbying at the state level, where platform firms may have greater influence than in any single municipality).

¹⁸² See *supra* Section II.C (reporting difference-in-differences and Callaway & Sant'Anna estimates of the VCA treatment effect on log hotel tax revenue).

¹⁸³ See *supra* Section II.C, tbl.2 (reporting that the Callaway & Sant'Anna estimator, which addresses potential bias from staggered treatment timing, produces an ATT estimate of 0.127, closely tracking the TWFE estimate of 0.141).

¹⁸⁴ Mary Riddel & Pavel Yakovlev, *Inferring Tax Compliance from Pass-Through: Evidence from Airbnb Tax Enforcement Agreements*, 103 REV. ECON. & STAT. 636, 636 (2021) (estimating that taxes were paid on at most twenty-four percent of Airbnb transactions before enforcement agreements took effect).

VCAs, they appear to do what they were designed to do: get cities paid.

The harder question is whether the revenue gains justify the concessions that accompanied them. The answer, I have argued, depends on context. For tourism-dependent communities with limited enforcement capacity and a single dominant platform, a VCA may genuinely be the best available instrument, and the empirical evidence suggests it will deliver real fiscal returns. But for cities whose primary challenge is the conversion of residential housing stock to unregulated commercial use, the data-access concessions embedded in most VCAs are not incidental costs—they are the central problem, because they foreclose the zoning, registration, and consumer safety enforcement that residents most urgently need.¹⁸⁵ The difficulty is that a VCA signed under the former set of conditions can become the latter as a city's short-term rental market evolves—a dynamic several cities in this study experienced firsthand. The decoupling of collection from reporting, the structural innovation at the heart of VCAs, is thus neither inherently good nor bad policy. It is a tool whose value depends entirely on what is given up in exchange, and on whether the government retains the capacity to renegotiate as circumstances change.

The lessons extend beyond Airbnb and hotel taxes. The same pattern—rapid platform growth, regulatory lag, bilateral negotiation from a position of municipal weakness—is likely to recur as platforms expand into other regulated industries.¹⁸⁶ The Airbnb experience reveals a domestic version of the capacity constraints that the public finance literature has long documented in developing economies: even wealthy democracies may find themselves adopting second-best tax instruments when their institutional infrastructure cannot keep pace with novel economic actors.¹⁸⁷ The strategies outlined in Part III—collective bargaining, cooperative monitoring, standardized data-sharing mandates, and selective state-level delegation—offer paths toward rebalancing this dynamic. Recent legislative developments in the European

¹⁸⁵ See, *supra* Section II.D (documenting cities' difficulties auditing aggregate remittances and enforcing zoning, registration, and consumer safety regulations without host-level data).

¹⁸⁶ *South Dakota v. Wayfair, Inc.*, 585 U.S. 162 (2018).

¹⁸⁷ See Timothy Besley & Torsten Persson, *Why Do Developing Countries Tax So Little?*, 28 J. ECON. PERSPS. 99 (2014).

Union and California suggest that the political conditions for such reforms may be ripening.¹⁸⁸ But the central insight is one of timing: the terms of a platform's civic obligations are far easier to establish before it becomes too entrenched to accept them. Cities that act early will negotiate from strength. Those that wait will find themselves, as so many did with Airbnb, choosing between an imperfect agreement and no agreement at all.

¹⁸⁸ See Council Regulation 2024/1028 of Apr. 11, 2024, on the Collection and Sharing of Data Relating to Short-Term Accommodation Rental Services, 2024 O.J. (L 1028) 1; S.B. 346, 2023–2024 Reg. Sess. (Cal. 2024) (effective Jan. 1, 2026).